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Automated XCT Data Analysis for Defect Detection in Additively Manufactured Metal Parts

*From NIST Benchmark Development to
PODFAM Industrial Validation*

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Note on this combined document

This document combines two things produced at different points in the same thesis project. Part I (Chapters 1–6 and Appendices A–B) is the master thesis mid-term report as submitted on April 26, 2026, covering method development and preliminary results on the NIST CoCr AM XCT benchmark dataset (`set1sample2raw`). Its content is reproduced essentially verbatim from that submission. Part II (Chapters 7–13) is the PODFAM validation phase chapter, completed after the mid-term and reporting full-volume results on real PODFAM process-monitoring XCT data, a separate piece of work using a different pseudo-labelling method (Bernsen instead of two-stage Otsu), added here as a continuation rather than a rewrite of Part I.

Three specific values in Part I were found, while preparing this combined document, to no longer match the working codebase’s actual configuration and have been corrected here rather than left silently wrong: the NLM denoising filter-strength factor (Section 3.3.4, Appendix B), the validation/test split percentages (Section 3.2.2, Appendix B), and the checkpoint output directory name (Appendix B). Each correction is marked in place. Appendix A’s repository structure has also been updated to reflect the codebase’s current layout, which has been substantially restructured (a `src/` package and a `scripts/` entry-point folder were introduced) since the mid-term submission; the original Appendix A tree described an earlier, now-superseded layout. Chapter 4’s figures (4.1–4.12) are retained as captioned placeholders — the specific NIST-phase source images referenced by those captions are not present in the current repository state and have not been regenerated for this document, since doing so was outside the scope of what was requested; the PODFAM phase’s own figures (Part II) are all freshly generated from the current pipeline and are embedded in full. All other content in Part I, including its reported quantitative results, is reproduced as originally submitted and was not re-verified against a pipeline re-run.

The bibliography at the end of Part I has been extended into a single, deduplicated reference list covering both parts; in-text citations within Part II use the same shared `references.bib` file as Part I, so every citation in this document resolves through one numbered list.

Summary

This master’s thesis develops and validates an automated pipeline for the detection of internal defects in metal additive manufacturing (AM) components using X-ray Computed Tomography (XCT) data. The work addresses a critical industrial challenge: the reliable and scalable identification of gas porosity, lack-of-fusion voids, and micro-cracks in high-resolution XCT volumes acquired from components fabricated by laser powder bed fusion (LPBF).

The primary dataset is the NIST CoCr AM XCT benchmark dataset (`set1sample2raw`), comprising 900 slices at 1010×980 pixel resolution with a raw intensity range of $[0, 59631]$. No expert-annotated ground truth masks are available for this dataset; defect detection labels are therefore generated automatically using a two-stage Otsu thresholding strategy applied within the metal component region only, followed by morphological cleaning and connected component filtering.

A four-stage preprocessing pipeline has been implemented and fully validated on the complete 900-slice dataset in 578.5 seconds on CPU. The pipeline comprises percentile normalisation, polynomial beam hardening correction (degree 3), polar-coordinate ring artefact suppression (radius 15 pixels), and non-local means denoising ($h = 0.6 \times \hat{\sigma}$, adaptive per slice — corrected from the mid-term submission’s stated value; see the Note on this combined document). The pipeline achieves a noise standard deviation reduction from 20952 to 0.385, a factor of $\times 54,500$, and an SNR improvement from 1.756 to 1.773. A systematic parameter sensitivity analysis across NLM filter strength, BHC polynomial degree, and ring filter radius confirms the selected configuration as the optimal operating point for this dataset.

Patch extraction (256×256 pixels, 50% overlap, stratified 1:3 foreground-to-background sampling) and a six-transform data augmentation pipeline have been implemented and validated on the NIST CoCr data, producing 15,840 patches at 64×64 and 3,860 patches at 128×128 from 20 sampled slices.

A 2D U-Net detection model has been implemented in PyTorch and initial training on the automatically generated labels is underway. After 20 training epochs, the combined Dice-Focal loss formulation achieves a validation Dice score of 0.68, compared to 0.41 for the standard binary cross-entropy baseline and 0.38 for the Otsu label generator itself. This confirms that the U-Net is already generalising beyond the capability of its own training labels, the fundamental goal of the pseudo-label approach.

The complete pipeline is version-controlled on GitHub, tracked with MLflow, and produces thesis-ready quantitative outputs including intensity histograms, SNR plots, parameter tuning comparisons, patch extraction grids, and augmentation validation figures.

This mid-term summary is extended, in Part II of this document, by a completed validation phase on real PODFAM metal AM XCT data: full-volume classical-threshold comparison across up to seven samples, a 2D U-Net trained on Bernsen pseudo-labels with defect-content-ranked sample selection (best validation Dice 0.600), held-out test evaluation, and an additional out-of-distribution inference check that surfaced and diagnosed a reproducible Bernsen calibration failure mode.

Preface

This thesis is carried out as a degree project for the Master of Engineering Science with a specialization in Robotics and Automation at the Department of Engineering Science, University West, Sweden. The work was conducted within the PODFAM research project during the spring semester of 2026, under the supervision of Amit Kumar Mishra and examination by Yongcui Mi.

The project was initiated with the goal of developing an automated pipeline for defect detection in additively manufactured (AM) metal components using X-ray computed tomography (XCT) data. For model development, I used benchmark datasets from the National Institute of Standards and Technology (NIST AM Bench), which provided diverse and well-annotated XCT scans suitable for training deep learning models. To ensure industrial relevance and robustness, the trained models were subsequently tested on PODFAM project datasets, consisting of XCT volumetric scans of AM metal parts collected within ongoing process monitoring research at University West.

I would like to express my sincere gratitude to my supervisor, Prof. Amit Kumar Mishra and Prof. Karthikeyan Thalavai Pandian for his continuous guidance, constructive feedback, and invaluable support throughout the project. His expertise in machine learning and industrial automation has been instrumental in shaping the direction of this work. I also extend my thanks to my examiner, Yongcui Mi, for her insightful comments and encouragement during the course of the thesis.

I am grateful to the PODFAM project team and collaborators for their engagement and for providing access to XCT data and domain expertise, which greatly enhanced the practical impact of this research. Special thanks also go to the Department of Engineering Science at University West for providing the resources, facilities, and academic environment that made this work possible.

Finally, I would like to thank my family and friends for their patience, encouragement, and unwavering support throughout my studies.

Affirmation

This master degree report, *Automated XCT Data Analysis for Defect Detection in Additively Manufactured Metal Parts*, was written as part of the master degree work needed to obtain a Master of Science with specialization in Robotics degree at University West. All material in this report that is not my own is clearly identified and used in an appropriate and correct way. The main part of the work included in this degree project has not previously been published or used for obtaining another degree.

Job George Konnoth Joseph

Date: April 26, 2026

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Symbols and glossary

AM	Additive Manufacturing — a process of building components layer by layer from metallic powder or wire feedstock using a laser or electron beam.
LPBF	Laser Powder Bed Fusion — the most common AM process, where a laser selectively melts successive layers of powder spread across a build platform.
DED	Directed Energy Deposition — an AM process using focused energy to melt material as it is deposited.
EBM	Electron Beam Melting — an AM process using an electron beam as the energy source.
XCT	X-ray Computed Tomography — a non-destructive imaging technique that reconstructs the three-dimensional internal structure of a component from a series of 2D X-ray projections.
NIST	National Institute of Standards and Technology — the US federal agency that provides the CoCr AM XCT benchmark dataset used for training and evaluation in this thesis.
CNN	Convolutional Neural Network — a deep learning architecture widely used for image analysis tasks, including defect detection in XCT data.
U-Net	A deep learning architecture with a symmetric encoder-decoder structure and skip connections, used in this thesis as the primary defect detection model.
NLM	Non-Local Means — a denoising algorithm that replaces each voxel intensity with a weighted average of similar voxels in a search window, preserving defect boundaries while suppressing noise.
BHC	Beam Hardening Correction — a preprocessing step that removes the systematic intensity gradient caused by differential X-ray attenuation in dense metal components.
SNR	Signal-to-Noise Ratio — a quantitative measure of image quality used to evaluate the effectiveness of each preprocessing stage.
GPU/CPU	Graphics Processing Unit / Central Processing Unit — hardware used for training and running deep learning defect detection models.
FAIR	Findable, Accessible, Interoperable, Reusable — principles for data stewardship ensuring long-term usability and reproducibility of datasets and pipeline outputs.
DVC	Data Version Control — a tool used alongside Git to version control large XCT datasets and ensure reproducibility of all experiments.
MLflow	An open-source platform used for experiment tracking, logging hyperparameters, metrics, and model checkpoints across all training runs.

LoF	Lack-of-Fusion — a type of internal defect in AM components caused by insufficient energy density during fabrication, resulting in irregularly shaped voids with sharp stress-concentrating features.
Gas Porosity	A type of internal defect in AM components formed by entrapped gas in the melt pool, typically spherical with diameters ranging from 1 to 500 micrometres.
Otsu Threshold	An automatic global thresholding method that maximises inter-class intensity variance. Used in this thesis as both a classical detection baseline and an automatic label generator.
Pseudo-Label	An automatically generated binary detection label produced by classical thresholding (Otsu for the NIST phase, Bernsen for the PODFAM phase), used as training supervision in the absence of expert-annotated ground truth.
Dice Coefficient	A detection accuracy metric measuring the overlap between predicted and reference defect regions, ranging from 0 (no overlap) to 1 (perfect overlap).
IoU	Intersection over Union — a detection accuracy metric measuring the ratio of overlap to union between predicted and reference defect regions.
BCE	Binary Cross-Entropy — a standard loss function for binary classification tasks. Shown to cause class collapse on severely imbalanced XCT defect data when used alone.
Dice-Focal Loss	The combined loss function used in this thesis, combining Dice loss and Focal loss with equal weighting to address severe class imbalance in XCT defect detection.
Patch Extraction	The process of dividing full XCT slices into fixed-size sub-images (256×256 pixels in this thesis) for input to the detection model.
ReLU	Rectified Linear Unit — the activation function used in the encoder and decoder stages of the U-Net, introducing non-linearity at low computational cost.
μm	Micrometre — unit of length used to describe defect sizes and XCT voxel dimensions. $1 \mu\text{m} = 10^{-6}$ metres.
10^6 K/s	Cooling rate in Kelvin per second, typical of rapid solidification dynamics in LPBF additive manufacturing processes.
Bernsen / DCT / LCT	Bernsen local adaptive thresholding, used as the primary pseudo-label

method for the PODFAM phase (Part II). DCT (dynamic contrast threshold) and LCT (local contrast) are its two governing parameters, defined in Section 9.3.

PART I
MID-TERM REPORT

NIST CoCr AM XCT Development Phase — as submitted April 26, 2026

1. Introduction

The global adoption of additive manufacturing for the production of structurally critical metal components has accelerated considerably over the past decade. Industries including aerospace, automotive, and biomedical engineering have embraced metal AM processes such as laser powder bed fusion (LPBF), directed energy deposition (DED), and electron beam melting (EBM) for their ability to produce components with complex internal geometries, reduced material waste, and shortened lead times compared to traditional machining and casting routes [1].

Despite these advantages, metal AM processes are inherently susceptible to the formation of internal defects during fabrication. These defects arise from the complex physics of rapid solidification, thermal gradients, powder feedstock variability, and laser or beam parameter instability. The most commonly observed defect classes include gas porosity arising from entrapped gas in the melt pool, lack-of-fusion (LoF) voids caused by insufficient energy density and incomplete melting of adjacent powder particles, micro-cracks resulting from thermal stresses and residual strain accumulation, and metallic inclusions originating from contaminated powder or redeposited spatter [2]. Each of these defect types can initiate fatigue crack growth under cyclic loading, dramatically reducing component service life and potentially causing catastrophic failure in safety-critical applications [3].

Effective, reliable, and efficient defect inspection is therefore not merely a quality assurance concern but a fundamental prerequisite for the safe deployment of AM components. X-ray computed tomography has established itself as the gold standard for volumetric non-destructive inspection, capable of imaging internal features at resolutions approaching the micrometre scale without requiring sectioning or destruction of the component [4]. The NIST CoCr AM XCT dataset used in this work, comprising 900 slices at 1010×980 pixel resolution with a raw intensity range of $[0, 59631]$, is representative of the scale and complexity typical of industrial XCT inspection tasks [5].

However, the volumetric nature of XCT data creates substantial challenges for manual analysis. Expert radiographers or materials engineers must review vast quantities of image data, a process that is time-consuming, costly, operator-dependent, and difficult to scale to industrial production volumes. The primary technical bottleneck in XCT-based defect analysis is not data acquisition but data interpretation: specifically, the automated detection and localisation of defect regions within noisy, artefact-affected volumetric images.

Classical approaches based on global thresholding such as Otsu’s method [6], morphological filtering, and edge detection fail under the imaging conditions typical of dense metal scans, where defects and the surrounding metal matrix occupy similar greyscale intensity ranges and beam hardening artefacts create systematic intensity gradients across the image volume [7]. In the NIST AM Bench dataset, Otsu thresholding achieves Dice scores of only

0.43–0.61 for porosity detection, compared to 0.82–0.91 for CNN-based approaches [8], confirming the limitations of classical methods for this application.

Deep learning, and in particular fully convolutional architectures such as U-Net [9], offers a fundamentally different approach. Rather than relying on hand-crafted intensity thresholds or morphological rules, deep learning models learn rich hierarchical representations of spatial context, texture, and shape directly from training data. This enables them to detect and localise defect voxels even under challenging noise and contrast conditions, and to generalise across different component geometries and scan settings when trained on sufficiently diverse datasets [2].

A critical practical challenge in this thesis is the absence of expert-annotated ground truth masks for the NIST CoCr AM XCT dataset [5]. This is addressed through an automatic label generation strategy: Otsu thresholding is applied within the metal region only, not the air background, to generate binary detection labels, which then serve as training supervision for the 2D U-Net. This approach follows Yosifov [10], who used Otsu-derived labels to train a modified 3D U-Net for XCT analysis of fibre-reinforced polymers, and is justified by the observation that the trained U-Net already achieves a validation Dice score of 0.68 after 20 epochs, compared to 0.41 for the BCE baseline and 0.38 for the Otsu label generator itself, demonstrating that the model learns to detect defects beyond the capability of its own training labels.

This thesis addresses the design, implementation, and empirical validation of a deep learning pipeline for automated XCT defect detection in additively manufactured metal parts. The complete pipeline, covering preprocessing, automatic label generation, patch extraction, data augmentation, model training, and evaluation, has been implemented in Python, version-controlled on Git, and validated on the NIST CoCr `set1sample2raw` dataset. Part II of this document extends the validation to real PODFAM process-monitoring data.

1.1 Aim

The aim of this thesis is to develop, implement, and validate an automated deep learning-based pipeline for the detection of internal defects in XCT scans of additively manufactured metal components. The work emphasises rigorous empirical methodology: systematic comparison of a deep learning detector (2D U-Net) against a classical baseline (Otsu thresholding) using standardised evaluation metrics on the NIST CoCr AM XCT benchmark dataset, followed by assessment of robustness across preprocessing configurations and loss function formulations.

The ultimate deliverable is a documented, validated, and open-source defect detection pipeline that can be adopted or extended by manufacturing organisations seeking to automate XCT-based quality assurance.

1.2 Research Questions

This thesis is structured around the following specific research questions:

1. **RQ1:** How effectively can a preprocessing pipeline reduce imaging artefacts and noise in XCT scans of AM metal components, as measured by signal-to-noise ratio

improvement and visual inspection of defect-relevant structures across all four pipeline stages?

2. **RQ2:** What preprocessing parameter configurations, specifically NLM filter strength, BHC polynomial degree, and ring artefact filter radius, produce optimal image quality for subsequent automated defect detection?
3. **RQ3:** How do preprocessing strategies, patch extraction, and data augmentation affect the quality of automatically generated detection labels and downstream 2D U-Net detection performance under conditions of severe class imbalance (1.2–2.8% defect voxels)?
4. **RQ4:** What loss function formulations best address the class imbalance inherent in XCT defect data, and how does loss function choice affect detection accuracy as measured by Dice coefficient, IoU, precision, and recall?
5. **RQ5:** Does a 2D U-Net trained on classically-derived pseudo-labels from one industrial dataset generalize to a fully held-out sample and to an out-of-distribution sample from the same process, as measured by both pixel-level overlap and volume-fraction agreement?
6. **RQ6:** Are classical thresholding methods (Otsu, Yen, Bernsen) reliable enough to serve as an independent, unsupervised comparison baseline across a real industrial sample set, or do they carry systematic, sample-dependent failure modes that could be mistaken for genuine measurement?

1.3 Contributions

The main contributions of this thesis are:

- A complete end-to-end Python pipeline for XCT defect detection covering preprocessing, automatic label generation, patch-based data preparation, 2D U-Net training, and evaluation, fully version-controlled on Git with MLflow experiment tracking and following FAIR data stewardship principles [11].
- A systematic preprocessing pipeline validated on the NIST CoCr `set1sample2raw` dataset (900 slices, 1010×980 pixels, raw range $[0, 59631]$), achieving noise standard deviation reduction from 20952 to 0.385, a factor of $\times 54,500$, and SNR improvement from 1.756 to 1.773 across the four-stage pipeline.
- A comprehensive preprocessing parameter sensitivity analysis evaluating NLM filter strength ($h \in \{0.5, 1.0, 1.5, 2.0\}$), BHC polynomial degree ($d \in \{1, 2, 3, 5\}$), and ring artefact filter radius ($r \in \{5, 10, 15, 25\}$ pixels), with quantitative SNR comparison across all configurations.
- An automatic label generation strategy using two-stage Otsu thresholding with morphological cleaning, enabling supervised U-Net training without manual annotation [6, 10].

- Preliminary empirical evidence that the 2D U-Net trained on automatic labels already outperforms the Otsu label generator itself (validation Dice 0.68 vs. 0.38 after 20 epochs), confirming that the model generalises beyond its noisy supervision signal.
- A documented, reproducible open-source implementation with thesis-ready analysis outputs including intensity histograms, SNR plots, parameter tuning comparisons, patch extraction grids, and augmentation validation figures.
- (Part II, answering RQ5) A completed validation of this pipeline on real PODFAM metal AM XCT data, including a trained detector evaluated on a fully held-out sample and an additional out-of-distribution sample.
- (Part II, answering RQ6) A documented and independently reproduced classical-thresholding calibration failure mode, traced to processed-image mean intensity and confirmed on two separate PODFAM samples rather than dismissed as a one-off anomaly.

2. Related Work (Background)

This chapter presents the theoretical foundations and related work underpinning the development of an automated defect detection pipeline for XCT data of additively manufactured metal components. It covers the physics of defect formation in AM processes, the principles of XCT imaging and its associated artefacts, classical and deep learning-based detection methods, and the mathematical foundations of the loss functions and evaluation metrics used in this thesis.

2.1 Additive Manufacturing and Internal Defect Formation

Metal additive manufacturing encompasses a family of layer-by-layer fabrication processes in which components are built from metallic powder or wire feedstock using a focused energy source [12]. The most industrially prevalent process is laser powder bed fusion (LPBF), in which a laser selectively melts successive layers of powder spread across a build platform. The rapid solidification dynamics, characterised by cooling rates on the order of 10^6 K/s, create complex microstructures and residual stress distributions that differ substantially from those produced by conventional casting [13].

Internal defects arise from several distinct physical mechanisms. Gas porosity forms when dissolved gas is entrapped in the melt pool; these defects are typically spherical with diameters ranging from a few micrometres to several hundred micrometres [13]. Lack-of-fusion (LoF) defects arise when adjacent scan tracks are incompletely fused due to insufficient energy density; they are irregularly shaped with sharp features that act as stress concentrators [12]. Micro-cracks can form due to solidification cracking or solid-state cracking driven by thermal residual stresses. Inclusions may arise from surface oxide films on powder particles, redeposited condensate, or contamination [1].

The mechanical consequences of these defects are severe. Even small pores or LoF voids can initiate fatigue crack growth under cyclic loading, dramatically reducing component service life. In safety-critical applications such as aerospace turbine blades or biomedical implants, undetected defects can lead to catastrophic failure [2]. This motivates the need for reliable, automated, and scalable defect detection methods.

2.2 X-ray Computed Tomography for Non-Destructive Inspection

X-ray computed tomography is a non-destructive imaging technique that reconstructs the three-dimensional internal structure of an object from a series of two-dimensional X-ray projection images acquired at multiple angles [4]. The achievable voxel size in an XCT system is governed by the geometric magnification relationship:

$$V_{xl} = \frac{\text{SOD}}{\text{SDD}} \times P_{xl} \quad (1)$$

where SOD is the source-to-object distance, SDD is the source-to-detector distance, and P_{xl} is the detector pixel pitch. For metallic AM components, typical XCT voxel sizes range from 1–50 μm [4].

The NIST CoCr AM XCT dataset used in this thesis provides a representative example of industrial-scale XCT data: 900 slices at 1010×980 pixel resolution, stored as 16-bit unsigned TIFF images with a raw intensity range of $[0, 59631]$ [5]. At this scale, manual inspection of all slices is impractical, motivating automated detection approaches.

2.2.1 Imaging Artefacts in Metal XCT

Several systematic imaging artefacts complicate automated defect detection in dense metal XCT scans [4]:

- **Beam hardening:** Differential attenuation of low-energy X-ray photons creates a cupping effect, a systematic intensity gradient where the component periphery appears brighter than the centre. This gradient can mask real defects near component boundaries or create false intensity variations that mimic defects.
- **Ring artefacts:** Non-uniformities in individual detector elements produce concentric ring patterns in reconstructed slices. These rings interfere with defect localisation by creating circular false-positive regions at fixed radial distances from the rotation centre.
- **Noise:** Quantum noise from photon counting statistics and electronic detector noise introduce random intensity fluctuations that reduce contrast between defect voids and the surrounding metal matrix.

Addressing these artefacts through preprocessing is a prerequisite for reliable automated defect detection, as classical and deep learning methods alike are sensitive to systematic intensity variations [7].

2.3 Preprocessing Methods for XCT Data

2.3.1 Percentile Normalisation

Raw XCT intensity values vary widely across scans and scanners due to differences in energy settings, detector gain, and material composition. Percentile normalisation clips

intensities to a robust range and rescales to $[0, 1]$:

$$x_{\text{norm}} = \frac{\text{clip}(x, p_1, p_{99}) - p_1}{p_{99} - p_1 + \varepsilon} \quad (2)$$

where p_1 and p_{99} are the 1st and 99th intensity percentiles and $\varepsilon = 10^{-7}$ prevents division by zero. This approach is robust to extreme outliers such as metal streak artefacts and hot pixels that would distort linear normalisation [7].

2.3.2 Beam Hardening Correction

Beam hardening correction estimates the systematic intensity trend along the beam axis by fitting a polynomial of degree d to the mean intensity profile and subtracting it [4]:

$$x_{\text{BHC}}[i] = x_{\text{norm}}[i] - (T(i) - \bar{T}), \quad T(i) = \sum_{k=0}^d a_k i^k \quad (3)$$

where $T(i)$ is the fitted polynomial trend and $\bar{T}$ is its mean, ensuring zero net intensity shift. A polynomial degree of $d = 3$ was selected through parameter sensitivity analysis on the NIST CoCr dataset, as lower degrees are insufficient to capture the non-linear cupping gradient while higher degrees risk over-fitting to slice-level intensity variations.

2.3.3 Ring Artefact Suppression

Ring artefacts appear as horizontal bands in the polar coordinate representation of each slice. The suppression algorithm converts each slice to polar coordinates, applies a median filter of radius r pixels along the angular axis, and converts back to Cartesian coordinates. A filter radius of $r = 15$ pixels was found optimal for the NIST CoCr dataset, achieving effective ring suppression without introducing spatial blurring that would obscure small defects.

2.3.4 Non-Local Means Denoising

Non-local means (NLM) denoising [14] replaces each voxel intensity with a weighted average of voxels in a search window, where weights are determined by patch-level intensity similarity:

$$\text{NLM}(x)_i = \sum_j w(i, j) x_j, \quad w(i, j) = \frac{\exp(-\|P_i - P_j\|^2/h^2)}{Z_i} \quad (4)$$

where P_i and P_j are intensity patches centred at voxels i and j , h is the filter strength parameter, and Z_i is a normalisation constant. Unlike Gaussian or median filters, NLM preserves sharp defect boundaries because it averages over structurally similar regions rather than spatially adjacent ones [14]. Applied to the NIST CoCr dataset with $h = 0.6 \times \hat{\sigma}$ (adaptive per slice; corrected from the mid-term submission's stated $1.0 \times \hat{\sigma}$, see the Note on this combined document), NLM reduces mean noise standard deviation from 20952 to 0.385, a factor of $\times 54,500$.

2.4 Classical Defect Detection Methods

2.4.1 Otsu Thresholding

The Otsu method [6] is the most widely used classical approach for XCT defect detection. Given a normalised grey-level histogram $h(i) = n_i/n$, the optimal threshold t^* maximises the inter-class variance:

$$\sigma_B^2(t) = \omega_1(t) \omega_2(t) [\mu_2(t) - \mu_1(t)]^2, \quad t^* = \arg \max_t \sigma_B^2(t) \quad (5)$$

where $\omega_1(t)$ and $\omega_2(t)$ are the class probabilities below and above threshold t , and $\mu_1(t), \mu_2(t)$ are the corresponding class means [6].

Otsu thresholding is fully automatic and computationally efficient, but fails under conditions typical of metal XCT: beam hardening creates systematic intensity gradients that shift the optimal threshold spatially, and low contrast between defect voids and the metal matrix limits separability [7]. On the NIST AM Bench dataset, Otsu achieves Dice scores of only 0.43–0.61 for porosity detection, compared to 0.82–0.91 for CNN-based approaches [8]. In this thesis, Otsu is used both as a classical detection baseline and as an automatic label generator for U-Net training supervision.

2.4.2 Morphological Post-Processing

Raw Otsu masks are refined through morphological operations to improve label quality [1]:

- Morphological opening (3×3 structuring element): removes isolated noise voxels that survive thresholding
- Morphological closing (3×3 structuring element): fills small holes within genuine defect regions
- Connected component filtering: discards components smaller than 5 voxels as noise artefacts

2.5 Deep Learning for Defect Detection

2.5.1 Convolutional Neural Networks

Deep convolutional neural networks (CNNs) have demonstrated transformative performance across image analysis tasks due to their translational equivariance and hierarchical feature learning [12]. A convolutional layer applies K learnable filters of spatial extent F to produce output feature maps. The output dimensions are:

$$H_2 = \frac{H_1 - F + 2P}{S} + 1, \quad W_2 = \frac{W_1 - F + 2P}{S} + 1 \quad (6)$$

where H_1, W_1 are input spatial dimensions, P is zero padding, and S is stride. The ReLU activation function introduces non-linearity at low computational cost:

$$\text{ReLU}(x) = \max(0, x) \quad (7)$$

2.5.2 U-Net Architecture

The U-Net architecture [9] was originally developed for biomedical image segmentation and has since become the dominant approach for defect detection in XCT data [1]. It consists of a symmetric encoder-decoder structure with skip connections that transfer spatial detail from encoder to decoder:

- **Encoder:** four downsampling stages, each comprising two 3×3 convolutional layers with batch normalisation and ReLU activation [9], followed by 2×2 max pooling. Feature channels double at each stage: $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$.
- **Bottleneck:** two 3×3 convolutional layers with 1024 channels and dropout regularisation ($p = 0.2$) to reduce overfitting on limited training data.
- **Decoder:** four upsampling stages with transposed convolutions and skip connection concatenation from the corresponding encoder level.
- **Output layer:** 1×1 convolution followed by sigmoid activation, producing per-voxel defect probability in $[0, 1]$:

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (8)$$

Skip connections are critical for detecting small defects, as they preserve high-resolution spatial detail that is lost during downsampling [9]. The 2D formulation processes individual XCT slices as 256×256 pixel patches, achieving within 2–4% of full 3D volumetric performance at 10–50 $\times$ lower computational cost [15].

2.5.3 Automatic Label Generation

A critical challenge in this thesis is the absence of expert-annotated ground truth masks for the NIST CoCr dataset. This is addressed through a two-stage automatic label generation strategy following Yosifov [10]:

1. A first-pass Otsu threshold separates the metal component from the air background, defining the region of interest
2. A second-pass Otsu threshold is applied within the metal region only, identifying dark voxels as defects

This two-stage approach is essential to avoid misclassifying the air background outside the cylindrical component as defects, a common failure mode of single-pass global thresholding on XCT data [6]. The resulting pseudo-labels are imperfect but sufficient to provide training supervision, with the U-Net learning to generalise beyond the label generator’s systematic errors [10].

2.6 Loss Functions for Class-Imbalanced Detection

A central challenge in XCT defect detection is severe class imbalance: defect voxels constitute only 1.2–2.8% of total scan volume [16]. Standard binary cross-entropy (BCE)

loss causes models to collapse to predicting background for all voxels, achieving high accuracy while detecting no defects:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N \left[g_i \log(p_i) + (1 - g_i) \log(1 - p_i) \right] \quad (9)$$

where p_i is the predicted defect probability and g_i is the ground truth label for voxel i .

2.6.1 Dice Loss

Dice loss directly optimises the overlap between predicted and true defect regions, making it inherently robust to class imbalance [16]:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i p_i g_i + \varepsilon}{\sum_i p_i + \sum_i g_i + \varepsilon} \quad (10)$$

where ε is a smoothing constant that prevents division by zero.

2.6.2 Focal Loss

Focal loss [17] addresses class imbalance by down-weighting easy background examples and focusing training on hard defect voxels:

$$\mathcal{L}_{\text{Focal}} = -\alpha(1 - p_t)^\gamma \log(p_t), \quad \alpha = 0.25, \gamma = 2 \quad (11)$$

where $(1 - p_t)^\gamma$ is the modulating factor that reduces the loss contribution of confidently classified background voxels.

2.6.3 Combined Dice-Focal Loss

The primary loss formulation used in this thesis combines Dice and Focal losses with equal weighting [16, 17]:

$$\mathcal{L}_{\text{DF}} = \lambda \mathcal{L}_{\text{Dice}} + (1 - \lambda) \mathcal{L}_{\text{Focal}}, \quad \lambda = 0.5 \quad (12)$$

This formulation benefits from Dice loss's class-imbalance robustness and Focal loss's hard-example mining, achieving a validation Dice score of 0.68 after 20 epochs compared to 0.41 for BCE alone [16].

2.7 Evaluation Metrics

Detection performance is quantified using the following standard metrics:

$$\text{DSC} = \frac{2|A \cap B|}{|A| + |B|} \quad (13)$$

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|} \quad (14)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (15)$$

where A is the predicted detection mask, B is the reference label, TP is true positives, FP is false positives, and FN is false negatives. The Dice coefficient [6] is the primary acceptance criterion, with a target of $DSC \geq 0.75$ on the held-out test set.

2.8 Data Augmentation

Data augmentation is essential for training robust defect detection models when labelled data is limited [7]. By applying random transformations to training patches, augmentation effectively increases dataset diversity and reduces overfitting to scan-specific characteristics. The augmentation operations applied in this thesis are summarised in Table 1.

Table 1: Data augmentation operations applied during training [7].

Augmentation	Probability	Parameters
Horizontal/vertical flip	0.5	—
Random 90° rotation	0.5	$k \in \{1, 2, 3\}$
Elastic deformation	0.3	$\alpha = 34, \sigma = 4$
Intensity scaling	0.5	Scale $\in [0.9, 1.1]$
Gaussian noise	0.5	$\sigma \in [0.01, 0.05]$
Gamma correction	0.5	$\gamma \in [0.8, 1.2]$

All transforms are applied identically to both image patch and corresponding label patch to maintain spatial alignment. Elastic deformation simulates the geometric variability of defect shapes across different scan positions, while intensity transforms simulate acquisition variability across different scan settings and materials [7].

2.9 Related Work Summary

Table 2 summarises the key findings from the literature that directly inform the design decisions in this thesis.

Table 2: Summary of directly relevant related work.

Study	Finding	Relevance to this thesis
Wong et al. [8]	CNN achieves Dice 0.82–0.91 vs. Otsu 0.43–0.61	Motivates U-Net over classical baseline
Bellens et al. [7]	Augmentation essential for limited labelled data	Justifies six-transform augmentation pipeline
Zubayer et al. [2]	Data diversity > architecture complexity	Supports 2D-first approach on diverse patches
Liu et al. [3]	Label quality is dominant performance factor	Justifies morphological cleaning of Otsu labels
Deshpande et al. [1]	Standardised benchmarks are the primary gap	Motivates use of NIST CoCr as benchmark dataset
Yosifov [10]	Otsu pseudo-labels enable U-Net training without annotation	Basis for automatic label generation strategy
Wong et al. [15]	2D within 2–4% of 3D at 10–50× lower cost	Justifies 2D slice-based detection approach
Sudre et al. [16]	Combined Dice-Focal loss best for class imbalance	Primary loss function selection

3. Method

This chapter describes the complete methodology developed for automated defect detection in XCT scans of additively manufactured metal components. The methodology covers the research design, dataset description, preprocessing pipeline, automatic label generation, patch extraction, data augmentation, detection model architecture, loss function selection, and evaluation framework. All implementation decisions are justified with reference to the theoretical background presented in Chapter 2.

3.1 Research Design

The research follows a mixed-methods design integrating systematic literature review, controlled empirical experimentation, and practical implementation evaluation [18]. Three sequential phases structure the work:

1. **Phase 1 — Literature Review and Framework Development:** Systematic review of deep learning for XCT defect detection, AM defect characterisation, preprocessing methods, and class imbalance mitigation strategies. This phase produces the research questions, methodology justification, and experimental design. *Status: complete.*
2. **Phase 2 — Controlled Empirical Experimentation:** Systematic implementation and benchmarking of the preprocessing pipeline, automatic label generation, and 2D U-Net defect detector on the NIST CoCr AM XCT benchmark dataset. Preprocessing validation and initial model training are complete. Full evaluation is in progress.
3. **Phase 3 — Validation and Pipeline Delivery:** Application of the validated pipeline to additional datasets for generalisation evaluation, manual annotation of a small test subset for independent ground truth validation, and delivery of the documented open-source pipeline. *Completed for the PODFAM dataset in Part II of this document.*

The empirical-comparative design is justified by the applied nature of the research problem. Performance differences between defect detection methods must be quantified under controlled conditions to provide actionable guidance for industrial adoption. A purely theoretical approach would not produce the validated performance estimates needed to assess industrial suitability [18].

3.2 Dataset

3.2.1 NIST CoCr AM XCT Dataset

The primary dataset used in this work is the NIST CoCr AM XCT dataset, publicly available from the NIST Intelligent Systems Division [5]. This dataset comprises high-resolution XCT scans of metal components fabricated by LPBF, provided as TIFF image stacks. The sample used is `set1sample2raw`, comprising 900 slices loaded and characterised as shown in Table 3.

Table 3: NIST CoCr AM XCT dataset properties measured from loaded TIFF slices (`set1sample2raw`).

Property	Value
Source	NIST Intelligent Systems Division [5]
Sample identifier	<code>set1sample2raw</code>
Number of slices	900
Slice resolution	1010 × 980 pixels
Data type	uint16 (16-bit unsigned)
Raw intensity range	[0, 59631]
Mean intensity	36763
Std deviation	20952
Preprocessed range	[0, 1] (float32)
Ground truth masks	Not available
Label strategy	Automatic (two-stage Otsu)

No expert-annotated defect masks are provided with this dataset. Automatic labels are generated using the Otsu-based pipeline described in Section 3.4. Full material and scan parameter specifications are available at the NIST Intelligent Systems Division website [5].

3.2.2 Data Split

The 900 preprocessed slices are divided into training, validation, and test sets using a stratified split to ensure proportional representation of defect-containing patches across all sets. The split actually implemented in the working codebase (`config.py`: `VAL_SPLIT`, `TEST_SPLIT`) is 70% training / 20% validation / 10% test; this corrects the mid-term submission’s originally stated 70/15/15 split, see the Note on this combined document:

- Training set: 70% of patches
- Validation set: 20% of patches (used for early stopping and hyperparameter selection)
- Test set: 10% of patches (held out until final evaluation)

Splitting is performed at the patch level after stratification by foreground/background label to ensure the 1:3 foreground-to-background ratio is preserved in all sets.

3.3 Preprocessing Pipeline

Raw XCT volumes are preprocessed through a four-stage sequential pipeline implemented in [data/loader.py](#). The pipeline is designed to remove systematic imaging artefacts that would otherwise interfere with reliable defect detection, while preserving the intensity characteristics of genuine defect structures. Figure 4.1 shows the visual effect of each stage on a representative NIST CoCr slice.

3.3.1 Stage 1 — Percentile Normalisation

Raw intensities are clipped to the 1st and 99th percentiles and rescaled to $[0, 1]$ per Equation (2), applied to the NIST CoCr `set1sample2raw` dataset with $p_1 = 0.0$, $p_{99} \approx 54146$, output range $[0.0000, 1.0000]$. This approach is robust to extreme outlier intensities such as metal streak artefacts and hot pixels [7].

3.3.2 Stage 2 — Beam Hardening Correction

A polynomial of degree $d = 3$ is fitted to the mean intensity profile along the depth axis and subtracted, per Equation (3). The polynomial degree $d = 3$ was selected through systematic parameter sensitivity analysis (Section 3.3.6). Processing time on the 900-slice NIST volume: 36.6 seconds.

3.3.3 Stage 3 — Ring Artefact Suppression

Each 2D slice is converted to polar coordinates centred on the component rotation axis. A median filter of radius $r = 15$ pixels is applied along the radial axis, where ring artefacts appear as systematic horizontal bands. The corrected data is then converted back to Cartesian coordinates. The radius $r = 15$ was selected through parameter sensitivity analysis as the minimum value achieving full ring suppression without spatial blurring of small defects. Processing time: 148.8 seconds for 900 slices.

3.3.4 Stage 4 — Non-Local Means Denoising

Non-local means denoising [14] is applied with parameters patch size 5, patch distance 6, and filter strength $h = 0.6 \times \hat{\sigma}$, where $\hat{\sigma}$ is the per-slice estimated noise standard deviation of each slice (corrected from the mid-term submission’s stated $h = 1.0 \times \hat{\sigma}$ to match the value actually implemented in `config.py`’s `NLM_H_FACTOR`; see the Note on this combined document), per Equation (4). Processing time: 368.3 seconds for 900 slices. Total pipeline time: 578.5 seconds (9.6 minutes on CPU) for the full 900-slice volume.

3.3.5 Preprocessing Results

Table 4 summarises the quantitative effect of the pipeline on the NIST CoCr dataset, measured from 50 sampled slices.

Table 4: Quantitative preprocessing results on NIST CoCr `set1sample2raw` (50 sampled slices). Noise reduction factor of $\times 54,500$. Pipeline time 578.5 s (900 slices, CPU).

Metric	Raw	Preprocessed
Intensity range	[0, 59631]	[0.0000, 1.0000]
p_{99} value	54146	1.0000
Mean SNR	1.756	1.773
Noise std dev	20952.2	0.385

3.3.6 Parameter Sensitivity Analysis

A systematic sensitivity analysis was conducted to select optimal values for the three key tunable parameters. Each parameter was varied independently while holding others fixed, and SNR was measured on the NIST CoCr dataset.

NLM filter strength $h \in \{0.5, 1.0, 1.5, 2.0\} \times \hat{\sigma}$: $h = 1.0$ selected as the optimal value for this sensitivity sweep; effective noise reduction without suppression of fine defect boundary sharpness. (Note: the value actually deployed in the trained pipeline is the adaptive factor 0.6, per Section 3.3.4 above; this sub-section reports the sweep as originally conducted and is not itself corrected.)

BHC polynomial degree $d \in \{1, 2, 3, 5\}$: degree 1 is insufficient for the non-linear cupping gradient; degree 3 provides effective correction; degree 5 risks over-correction near volume boundaries. Degree 3 selected.

Ring filter radius $r \in \{5, 10, 15, 25\}$ pixels: radius 5 is insufficient; radius 15 achieves full suppression without spatial blurring; radius 25 over-smooths in the radial direction. Radius 15 selected.

3.4 Automatic Label Generation

Since no expert-annotated ground truth masks are available for the NIST CoCr dataset, binary detection labels are generated automatically using a two-stage Otsu thresholding pipeline implemented in [data/pseudo_labels.py](#), following the approach of Yosifov [10]:

1. **Stage 1 — Part boundary detection:** A first-pass Otsu threshold is applied to the full slice to separate the metal component from the air background. This defines the region of interest (ROI) inside the cylindrical part boundary. Single-pass global thresholding applied to the full field of view incorrectly labels the air background as defects; the two-stage approach eliminates this failure mode [6].
2. **Stage 2 — Defect detection within ROI:** A second Otsu threshold is applied within the metal ROI only, identifying dark voxels below the within-metal threshold as defects.
3. Morphological opening (3×3 structuring element): removes isolated noise voxels that survive thresholding.
4. Morphological closing (3×3 structuring element): fills small holes within genuine defect regions.

5. Connected component filtering: discards components smaller than 5 voxels as noise artefacts.

The resulting pseudo-labels are imperfect but sufficient to provide training supervision. Preliminary results show that the U-Net already achieves a validation Dice score of 0.68 after 20 epochs, surpassing the Otsu label generator itself (0.38), confirming that the model learns to detect defects beyond the capability of its training labels [10].

3.5 Patch Extraction

Full XCT volumes are divided into fixed-size 2D patches for model input, implemented in [data/dataset.py](#). The patch extraction configuration is:

- Patch size: 256×256 pixels
- Stride: 128 pixels (50% overlap between adjacent patches)
- Minimum defect pixels for a foreground patch classification: 10
- Foreground-to-background patch ratio: 1:3 (stratified sampling)

Random patch extraction would yield $> 95\%$ background-only patches due to the severe class imbalance (1.2–2.8% defect voxels). The stratified 1:3 foreground-to-background oversampling strategy ensures sufficient gradient signal from the minority defect class during training [16].

From 20 uniformly sampled slices of the NIST CoCr preprocessed volume, the following patch counts were extracted at different patch sizes for validation purposes:

Table 5: Patch counts extracted from 20 sampled NIST CoCr slices with 50% overlap stride.

Patch size	Total patches
64×64	15,840
128×128	3,860
256×256	Used for model training

3.6 Data Augmentation

Data augmentation is applied online during training to increase dataset diversity and reduce overfitting to scan-specific characteristics [7]. All transforms are applied identically to both image patch and corresponding label patch to maintain spatial alignment. The augmentation pipeline is implemented in [data/augmentation.py](#) and validated on NIST CoCr patches as shown in Figure 4.12 (same operations as Table 1, Section 2.8).

3.7 Detection Model Architecture

3.7.1 2D U-Net

The primary detection model is a 2D U-Net [9] implemented in PyTorch [19] ([models/unet2d.py](#)). The 2D formulation processes individual XCT slices as 256×256 pixel patches, achieving within 2–4% of full 3D volumetric performance at $10\text{--}50\times$ lower computational cost [15]. The architecture configuration is:

- Encoder: four downsampling stages, each comprising two 3×3 convolutional layers with batch normalisation and ReLU activation [9], followed by 2×2 max pooling. Feature channels double at each stage: $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$.
- Bottleneck: two 3×3 convolutional layers with 1024 feature channels and dropout regularisation ($p = 0.2$).
- Decoder: four upsampling stages with transposed convolutions and skip connection concatenation from the corresponding encoder levels.
- Output layer: 1×1 convolution with sigmoid activation, producing per-voxel defect probability in $[0, 1]$.
- Trainable parameters: approximately 31 million.
- Optimiser: Adam [20] with learning rate 4×10^{-5} , trained for up to 50 epochs with early stopping (patience 10 epochs).

3.7.2 Classical Baseline

Otsu thresholding with two-stage part-boundary detection and morphological post-processing (Section 3.4) serves as the classical lower-bound baseline, enabling direct quantitative comparison of deep learning detection performance against the automatic label generator itself.

3.8 Loss Functions

Four loss function formulations are evaluated to determine the most effective approach for defect detection under severe class imbalance, implemented in [training/losses.py](#):

1. Binary cross-entropy (BCE): standard baseline, Equation (9). Known to cause class collapse on severely imbalanced XCT data [16].
2. Dice loss: directly optimises detection overlap, inherently robust to class imbalance [16], Equation (10).
3. Focal loss: down-weights easy background examples, focuses training on hard defect voxels [17], Equation (11).
4. Combined Dice-Focal (primary formulation) [16, 17], Equation (12).

Preliminary results confirm that the combined Dice-Focal loss achieves a validation Dice score of 0.68 after 20 epochs, compared to 0.41 for BCE alone, confirming its superiority for class-imbalanced defect detection [16].

3.9 Evaluation Metrics

All models are evaluated using the metrics of Equations (13)–(15), implemented in [training/metrics.py](#), where A is the predicted detection mask, B is the reference label, and TP, FP, FN denote true positives, false positives, and false negatives at the voxel level.

3.9.1 Acceptance Criteria

- Dice coefficient ≥ 0.75 on held-out test set (primary criterion)
- IoU ≥ 0.60 on held-out test set
- Recall ≥ 0.80 for defects with diameter $> 50 \mu\text{m}$
- Inference time ≤ 120 seconds per full scan volume on a single GPU

3.10 Software Implementation

The complete pipeline is implemented as a modular Python package version-controlled on Git:

- Language: Python 3.10
- Deep learning: PyTorch [19] with CUDA support
- Image processing: NumPy, scikit-image, SciPy for volume handling and preprocessing
- Experiment tracking: MLflow for logging hyperparameters, metrics, and model checkpoints across all training runs
- Version control: Git with DVC for dataset versioning, ensuring full reproducibility of all experiments
- Visualisation: Matplotlib for quantitative analysis outputs, Plotly for interactive 3D defect maps

3.11 Experimental Design Summary

Table 6: Summary of experimental phases, methods, and acceptance criteria. Phase 3A/3B status updated to reflect Part II’s completion; all other statuses reproduced as of the April 2026 mid-term submission.

Phase	Objective	Method	Status
1	Literature synthesis	Systematic review	Complete
2A	Preprocessing validation	Four-stage pipeline on NIST CoCr	Complete
2B	Parameter sensitivity	NLM, BHC, ring filter tuning	Complete
2C	Label generation	Two-stage Otsu + morphological cleaning	In progress
2D	Baseline detection	Otsu vs. 2D U-Net	In progress
2E	Loss function evaluation	BCE vs. Dice vs. Focal vs. combined	Planned
3A	Generalisation evaluation	Additional datasets (PODFAM)	Complete (Part II)
3B	Pipeline delivery	Documented Git repository	Complete (Part II)

4. Work

This chapter documents all work completed to date, the current project status, and planned next steps. All preprocessing results presented here were produced by running the implemented pipeline on the NIST CoCr `set1sample2raw` dataset (900 slices, 1010×980 pixels, raw intensity range $[0, 59631]$).

4.1 Completed Activities

4.1.1 Literature Review

A comprehensive literature review was conducted covering deep learning architectures for XCT defect detection, AM defect characterisation and formation mechanisms, preprocessing methods for volumetric XCT data, class imbalance mitigation strategies, and loss function design for imbalanced segmentation. Approximately 35 peer-reviewed papers were reviewed and synthesised. Key findings are presented in Chapter 2.

4.1.2 Dataset Acquisition and Characterisation

The NIST CoCr AM XCT dataset (`set1sample2raw`) was downloaded from the NIST Intelligent Systems Division and characterised from loaded TIFF slices. Measured properties: 900 slices, 1010×980 pixel resolution, uint16 data type, raw intensity range $[0, 59631]$, mean intensity 36763, standard deviation 20952. No expert-annotated ground truth masks are provided; automatic label generation is used as described in Chapter 3.

Initial exploratory data analysis confirmed severe class imbalance (approximately 1.2–2.8% defect voxels across sampled slices), consistent with published values for AM XCT porosity data [16].

4.1.3 Preprocessing Pipeline

The complete four-stage preprocessing pipeline was implemented in `data/loader.py` and validated on the full NIST CoCr `set1sample2raw` dataset. All 900 slices were processed successfully with a total pipeline time of 578.5 seconds on CPU.

Figure 4.1: Visual comparison of all five preprocessing stages on a representative NIST CoCr `set1sample2raw` slice. From left to right: raw input (range $[0, 59631]$), after percentile normalisation, after beam hardening correction, after ring artefact suppression, and after NLM denoising.

[Figure 4.1 — Preprocessing stage comparison]

Source image (raw / normalised / BHC / ring-suppressed / NLM-denoised panel for a NIST set1sample2raw slice) not present in the current repository; not regenerated for this document.

[Figure 4.2 — ROI zoom across preprocessing stages]

Source image not present in the current repository; not regenerated for this document.

Figure 4.2: Zoomed region of interest (centre quarter) across all preprocessing stages. Ring artefact suppression and NLM smoothing effects are clearly visible at the sub-millimetre scale where small defects are located.

4.1.4 Statistical Analysis of Preprocessing Effects

Quantitative SNR and noise measurements were computed from 50 sampled slices before and after preprocessing. Mean SNR improved from 1.756 to 1.773 and mean noise standard deviation reduced from 20952 to 0.385, a reduction factor of $\times 54,500$, confirming the effectiveness of the pipeline.

[Figure 4.3 — Intensity histograms]

Source image not present in the current repository.

Figure 4.3: Intensity histograms computed from 10 sampled NIST CoCr slices. Left: raw 16-bit intensities (range $[0, 59631]$). Right: normalised preprocessed intensities (range $[0, 1]$).

[Figure 4.4 — Per-slice SNR and noise std]

Source image not present in the current repository.

Figure 4.4: Per-slice SNR (left) and noise standard deviation (right) for raw (red) and preprocessed (blue) NIST CoCr data across 50 sampled slices.

Figure 4.5: Statistical summary comparing raw and preprocessed NIST CoCr volume metrics across 50 sampled slices.

4.1.5 Parameter Sensitivity Analysis

A systematic parameter sensitivity analysis was conducted on the NIST CoCr dataset to select optimal values for the three key tunable preprocessing parameters.

[Figure 4.5 — Statistical summary table]

Source image not present in the current repository.

[Figure 4.6 — NLM filter strength comparison]

Source image not present in the current repository.

Figure 4.6: NLM filter strength comparison ($h \in \{0.5, 1.0, 1.5, 2.0\} \times \hat{\sigma}$) on the central NIST CoCr slice.

[Figure 4.7 — BHC polynomial degree comparison]

Source image not present in the current repository.

Figure 4.7: Beam hardening correction polynomial degree comparison ($d \in \{1, 2, 3, 5\}$).

Figure 4.8: Ring artefact filter radius comparison ($r \in \{5, 10, 15, 25\}$ pixels).

Figure 4.9: Combined SNR comparison across NLM h -factor and BHC polynomial degree settings.

[Figure 4.8 — Ring filter radius comparison]

Source image not present in the current repository.

[Figure 4.9 — Combined SNR comparison]

Source image not present in the current repository.

4.1.6 Patch Extraction

The patch extraction pipeline was implemented and validated on 20 uniformly sampled NIST CoCr preprocessed slices. Patches with standard deviation below 0.01 were filtered as uninformative background.

[Figure 4.10 — Representative patches]

Source image not present in the current repository.

Figure 4.10: Representative 128×128 pixel patches extracted from 20 sampled NIST CoCr slices with 50% overlap stride.

Figure 4.11: Total patch counts at 64×64 (15,840 patches) and 128×128 (3,860 patches) sizes from 20 sampled slices with 50% overlap.

[Figure 4.11 — Patch counts]*Source image not present in the current repository.*

4.1.7 Data Augmentation Validation

All six augmentation transforms were validated on extracted NIST CoCr patches. Correct image-label spatial alignment was confirmed for all transforms.

[Figure 4.12 — Augmentation pipeline validation]*Source image not present in the current repository.*

Figure 4.12: Augmentation pipeline validation on a representative NIST CoCr patch (128×128 pixels). Original patch and eight independent augmentations produced by the full pipeline [7].

4.1.8 Software Pipeline

The complete software pipeline has been implemented and committed to a structured Git repository with MLflow experiment tracking. Its module layout has since evolved; see Appendix A for the current, corrected repository structure.

4.1.9 Initial Detection Model Training

Initial 2D U-Net training on NIST CoCr data is underway. After 20 training epochs with the combined Dice-Focal loss, a validation Dice score of 0.68 has been achieved, compared to 0.41 for the BCE baseline and 0.38 for the Otsu label generator itself. This confirms that the model is already learning to detect defects beyond the capability of its own training labels [10].

4.2 Current Status

The project is currently in the active model training and evaluation phase (Phase 2C–2D per Table 6). The preprocessing pipeline has been fully validated on the NIST CoCr dataset with quantitative SNR measurements, parameter sensitivity analysis, and visual confirmation all completed. Automatic label generation and 2D U-Net training are underway. Part II of this document reports the subsequent completion of Phase 3A/3B on the PODFAM dataset.

4.3 Planned Next Steps

- Complete 2D U-Net training and full quantitative evaluation on NIST CoCr held-out test set (target: $\text{Dice} \geq 0.75$)
- Compare all four loss function formulations (BCE, Dice, Focal, combined Dice-Focal) systematically
- Assess the independent contribution of each augmentation transform to detection performance
- Manually annotate 5–10 representative NIST CoCr slices in ITK-SNAP to provide independent ground truth for final validation
- Apply pipeline to additional datasets for cross-material generalisation evaluation (completed for PODFAM in Part II)
- Begin Phase 3 industrial validation pipeline development
- Draft full results and discussion chapter with quantitative comparison tables and detection visualisations

5. Preliminary Results and Discussion

This chapter presents the preliminary results obtained from the preprocessing pipeline validation, statistical analysis, parameter sensitivity analysis, and initial detection model training on the NIST CoCr `set1sample2raw` dataset. All quantitative results reported here are measured from the implemented pipeline as described in Chapter 3.

5.1 Key Insights from Literature Review

The systematic literature review produced several findings with direct implications for the experimental design of this thesis:

- U-Net-based deep learning consistently outperforms classical threshold-based detection methods by 20–40 Dice percentage points on AM XCT data, with the largest gains observed for small defects and artefact-affected regions [8].
- 2D slice-based detection achieves within 2–4% of full 3D volumetric performance on isotropic defect types such as gas porosity, at $10\text{--}50\times$ lower computational cost [15]. This finding directly motivates the 2D-first approach adopted in this thesis.
- Combined Dice-Focal loss achieves superior detection performance over either component alone under the class imbalance conditions typical of AM XCT data [7]. Preliminary training results on NIST CoCr data confirm this finding (Section 5.5).
- Increasing architecture complexity yields diminishing returns unless training data diversity is simultaneously increased [2]. This supports the data-centric approach adopted in this work, prioritising preprocessing quality and augmentation diversity over architectural novelty.
- Preprocessing, particularly beam hardening correction and percentile normalisation, substantially affects detection performance and is frequently under-reported in published studies. This thesis addresses this gap with a comprehensive parameter sensitivity analysis (Section 5.2).
- Training label quality is the dominant performance factor, exceeding the contribution of architecture choice [3]. This justifies the morphological cleaning applied to Otsu pseudo-labels in this work.

5.2 Preprocessing Pipeline Results

5.2.1 Visual Validation

The four-stage preprocessing pipeline was successfully applied to the full NIST CoCr `set1sample2raw` dataset (900 slices, 1010×980 pixels) in 578.5 seconds total on CPU. Figure 4.1 shows the visual effect of each stage on a representative central slice. The raw slice exhibits the characteristic high-intensity cylindrical cross-section with visible ring artefacts and beam hardening gradient. Each successive stage progressively reduces these artefacts while preserving the internal defect structures visible as dark spots within the metal.

Figure 4.2 confirms that fine defect boundary sharpness is preserved after NLM denoising, a critical requirement for reliable detection of sub-100 μm defects.

5.2.2 Quantitative Results

Figure 4.3 shows intensity histograms before and after preprocessing. The raw bimodal distribution (a background peak near zero and a broad metal peak) is compressed to a clean normalised distribution in $[0, 1]$ with clear separation between background, metal matrix, and defect intensity regions. This separation is a prerequisite for reliable automatic label generation.

Figure 4.4 presents per-slice SNR and noise standard deviation measurements. Table 7 summarises the key quantitative outcomes.

Table 7: Quantitative preprocessing results on NIST CoCr `set1sample2raw` (50 sampled slices from 900 total). Noise reduction factor of $\times 54,500$. Total time 578.5 s (900 slices, CPU).

Metric	Raw	Preprocessed
Intensity range	[0, 59631]	[0.0000, 1.0000]
p_{99} value	54146	1.0000
Mean SNR	1.756	1.773
Noise std dev	20952.2	0.385

The dominant contribution to noise reduction comes from the NLM denoising stage (368.3 seconds), while beam hardening correction and ring suppression primarily address systematic artefacts that affect defect localisation accuracy rather than raw noise level.

5.2.3 Parameter Sensitivity Analysis Results

Figures 4.6–4.9 present the results of the systematic parameter sensitivity analysis. The selected optimal configuration is $h = 1.0$ (sweep value; the deployed pipeline uses the adaptive factor 0.6, Section 3.3.4), polynomial degree $d = 3$, and ring filter radius $r = 15$ pixels.

Key observations from the sensitivity analysis:

- NLM filter strength: $h = 0.5$ leaves residual noise visible as speckle; $h = 1.5$ and above begin to smooth fine defect boundaries. $h = 1.0$ provides the best balance between noise suppression and defect boundary preservation within this sweep.

- BHC polynomial degree: Degree 1 is insufficient, a linear trend cannot capture the non-linear cupping gradient visible across the component diameter. Degree 3 fully corrects the gradient. Degree 5 shows marginal further improvement but introduces risk of over-correction in thin boundary regions.
- Ring filter radius: Radius 5 suppresses only the narrowest rings; dominant ring artefacts remain visible. Radius 15 achieves full suppression. Radius 25 begins to blur radial structures that could be confused with fine defect features.

5.3 Patch Extraction and Augmentation Results

5.3.1 Patch Extraction

Figure 4.10 shows a representative grid of 128×128 patches extracted from 20 sampled NIST CoCr preprocessed slices. Internal defects are clearly visible as dark spots in several patches, confirming that the preprocessing pipeline preserves defect contrast and that patch extraction correctly captures defect-containing regions.

Figure 4.11 shows total patch counts at two patch sizes: 15,840 patches at 64×64 and 3,860 patches at 128×128 . For model training, 256×256 patches are used with a stratified 1:3 foreground-to-background ratio to address class imbalance.

5.3.2 Augmentation Validation

Figure 4.12 shows the original patch and eight independently augmented variants produced by the full augmentation pipeline. All six transform types produce visually plausible variations in defect appearance, scan contrast, and component orientation. Spatial alignment between image and label patches was verified for all transforms by visual inspection.

5.4 Exploratory Dataset Analysis

Initial analysis of the NIST CoCr `set1sample2raw` dataset confirms the challenges documented in the literature:

- Class imbalance: defect voxels constitute approximately 1.2–2.8% of total scan volume across sampled slices, confirming that specialised loss function design is essential for reliable detection [16].
- Defect size distribution: the majority of detectable defects (approximately 68%) have equivalent spherical diameter $< 100 \mu\text{m}$, placing them in the range where detection is most challenging and where the U-Net’s skip connections, preserving high-resolution spatial detail, are most critical [9].
- Beam hardening artefacts: visible in all slices near the component boundary, generating systematic intensity gradients of approximately 8–12% of peak intensity range. BHC successfully removes these gradients as confirmed in Figure 4.1.
- Ring artefacts: present across sampled slices at low to moderate severity, validating the inclusion of ring artefact suppression in the preprocessing pipeline.

5.5 Preliminary Detection Model Results

Initial 2D U-Net training on NIST CoCr automatic labels is underway. After 20 training epochs the following results have been observed:

Table 8: Preliminary detection results after 20 training epochs on NIST CoCr set1sample2raw validation set.

Method	Validation Dice	Status
Otsu baseline (label generator)	0.38	Classical baseline
2D U-Net + BCE loss	0.41	Training complete
2D U-Net + Dice-Focal loss	0.68	Training in progress
Target acceptance criterion	0.75	To be achieved

The combined Dice-Focal loss already demonstrates a 66% improvement over BCE (0.68 vs. 0.41) and, critically, already surpasses the Otsu label generator itself (0.68 vs. 0.38), demonstrating that the U-Net is generalising beyond its noisy supervision signal [10]. Full training and evaluation against the held-out test set is planned in the next phase.

5.6 Challenges Identified

- **No ground truth for independent validation:** Since pseudo-labels are derived from Otsu thresholding, Dice scores computed against pseudo-labels measure agreement with Otsu rather than true defect detection accuracy. Manual annotation of 5–10 representative slices is planned to provide independent ground truth for final validation.
- **Data imbalance at patch level:** Random patch extraction yields greater than 95% background-only patches. The implemented stratified 1:3 foreground-to-background sampling strategy addresses this, verified on NIST CoCr data before deployment.
- **NLM processing time:** The NLM denoising stage requires 368.3 seconds for 900 slices on CPU. GPU-accelerated NLM or a faster denoising alternative (bilateral filter) will be evaluated if processing time becomes a bottleneck in the full training pipeline.
- **Pseudo-label quality near boundaries:** Otsu pseudo-labels under-detect LoF defects in beam-hardening-affected regions near component boundaries. The BHC stage mitigates this, and the U-Net is expected to partially correct these systematic label errors through spatial context learning [10].

6. Conclusion

This mid-term report documents the progress of a master’s thesis project on automated XCT defect detection in additively manufactured metal components using deep learning. The project is on schedule and has completed the literature review, research design, dataset acquisition and characterisation, full preprocessing pipeline implementation and validation, patch extraction, data augmentation, parameter sensitivity analysis, and initial detection model training on the NIST CoCr `set1sample2raw` dataset.

6.1 Preprocessing Pipeline

The four-stage preprocessing pipeline, percentile normalisation, beam hardening correction, ring artefact suppression, and non-local means denoising, was successfully validated on the full NIST CoCr `set1sample2raw` dataset (900 slices, 1010×980 pixels, raw intensity range $[0, 59631]$) in 578.5 seconds on CPU. The pipeline achieves a noise standard deviation reduction from 20952 to 0.385, a factor of $\times 54,500$, and an SNR improvement from 1.756 to 1.773. Visual confirmation (Figures 4.1 and 4.2) demonstrates that defect boundary sharpness is preserved throughout all preprocessing stages, which is essential for reliable detection of sub-100 μm defects.

Parameter sensitivity analysis (Figures 4.6, 4.7, 4.8, 4.9) confirmed the optimal configuration: NLM filter strength $h = 1.0$ (sweep value; deployed pipeline uses adaptive 0.6), BHC polynomial degree $d = 3$, and ring filter radius $r = 15$ pixels. This represents an original empirical contribution, preprocessing parameter selection is frequently omitted or unjustified in published XCT defect detection studies.

6.2 Patch Extraction and Augmentation

Patch extraction with 256×256 pixel patches and 50% overlap stride was validated on NIST CoCr data, producing 15,840 patches at 64×64 and 3,860 patches at 128×128 from 20 sampled slices (Figures 4.10 and 4.11). The stratified 1:3 foreground-to-background sampling strategy was implemented and verified to address the severe class imbalance of 1.2–2.8% defect voxels [16].

All six augmentation transforms were validated on NIST CoCr patches (Figure 4.12), confirming correct spatial alignment between image and label patches across all transform types [7].

6.3 Preliminary Detection Results

Initial 2D U-Net training on NIST CoCr automatic labels demonstrates clear improvement over both the classical baseline and the BCE loss formulation:

Table 9: Preliminary detection results after 20 training epochs.

Method	Validation Dice	vs. Target
Otsu baseline	0.38	—
2D U-Net + BCE	0.41	−45% below target
2D U-Net + Dice-Focal	0.68	−9% below target
Acceptance criterion	0.75	Target

The combined Dice-Focal loss already surpasses the Otsu label generator itself (0.68 vs. 0.38), confirming that the U-Net is learning to detect defects beyond the capability of its training labels [10]. Full training to convergence and evaluation against the held-out test set is the immediate next priority.

6.4 Key Conclusions

- **Preprocessing is essential and quantifiable:** The four-stage pipeline reduces noise by a factor of $\times 54,500$ and removes systematic artefacts that would otherwise mask or mimic defects. Without preprocessing, reliable automatic label generation and deep learning detection are not feasible on this dataset.
- **Automatic label generation is viable:** The two-stage Otsu strategy, applying thresholding within the metal region only, avoids the background misclassification failure mode of single-pass global thresholding. The U-Net already generalises beyond its noisy labels, validating this approach [6, 10].
- **Loss function choice is critical:** Combined Dice-Focal loss achieves 66% higher validation Dice than BCE (0.68 vs. 0.41) after 20 epochs, confirming that class-imbalance-aware loss functions are essential for XCT defect detection [16, 17].
- **2D detection is practical and competitive:** The 2D U-Net processes 256×256 patches efficiently, consistent with published findings that 2D approaches achieve within 2–4% of full 3D performance at substantially lower computational cost [15].
- **Standardised benchmarking enables reproducibility:** The use of the publicly available NIST CoCr AM XCT dataset [5] ensures that all results can be independently reproduced and compared against future work.
- **An end-to-end documented pipeline is the primary deliverable:** The complete pipeline, from raw TIFF stack to trained detector with MLflow tracking and DVC dataset versioning, is version-controlled on Git and designed for adoption by the AM inspection research community.

6.5 Remaining Work

The following work remained after the mid-term submission; Part II of this document reports what has since been completed:

- Complete 2D U-Net training to convergence and full evaluation on the held-out NIST CoCr test set against all four acceptance criteria (Dice ≥ 0.75 , IoU ≥ 0.60 , Recall ≥ 0.80 for defects $> 50\mu\text{m}$, inference time ≤ 120 seconds)
- Systematic comparison of all four loss function formulations (BCE, Dice, Focal, combined Dice-Focal)
- Independent assessment of each augmentation transform’s contribution to detection performance
- Manual annotation of 5–10 representative NIST CoCr slices for independent ground truth validation
- Application to additional datasets for cross-material generalisation evaluation, completed for PODFAM data in Part II
- Full results, discussion, and conclusion chapter, completed for the PODFAM phase in Part II
- Final pipeline documentation and open-source release on Git

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Appendix A — Software Repository and Pipeline

This appendix documents the software repository structure, installation instructions, and pipeline execution steps for the automated XCT defect detection pipeline developed in this thesis. The repository structure below reflects the codebase’s current, actual layout as of August 2026, which has been substantially restructured since the April 2026 mid-term submission (a [src/](#) package and a [scripts/](#) entry-point folder were introduced, and preprocessing/label-generation logic moved out of the originally-described [data/loader.py](#) and [data/pseudo_labels.py](#)); see the Note on this combined document.

A.1 Repository Access

Table 10: Repository information.

Field	Value
Platform	GitHub
Repository URL	https://github.com/jobgkj/podfam_research_project_XCT_Analysis
Branch	main
Language	Python 3.10+
Last commit	August 2026

A.2 Repository Structure (current)

Top level:

- `config.py` — all hyperparameters and paths, including PODFAM-phase additions (TRAINING_SAMPLES_OVERRIDE, MAX_TRAINING_SAMPLES, EXPERIMENT_NAME, Bernsen parameters)
- `pipeline.py` — end-to-end 2D(+3D) training entry point

[src/](#) — preprocessing, thresholding, and I/O (introduced after the mid-term; supersedes the originally-planned [data/loader.py](#) and [data/pseudo_labels.py](#)):

- [src/preprocess.py](#) — per-slice preprocessing (median filter + adaptive NLM denoising)

- [src/thresholding.py](#) — Otsu, Yen, and Bernsen thresholding (Bernsen is the PODFAM-phase primary pseudo-label source; see Part II, Section 9.3)
- [src/sample_mask.py](#) — circular sample-boundary detection
- [src/io.py](#) — mask generation orchestration
- [src/metrics.py](#) — per-pore connected-component statistics

[data/](#) — dataset and caching:

- [data/cache.py](#) — memory-mapped volume/mask cache (introduced after the mid-term, for training on volumes too large to hold fully in RAM)
- [data/dataset.py](#), [data/dataset_3d.py](#) — PyTorch Dataset and patch extraction
- [data/augmentation.py](#) — online augmentation transforms

[models/](#):

- [models/unet2d.py](#) — 2D U-Net architecture
- [models/unet3d.py](#) — 3D U-Net architecture
- [models/thesis_analysis.py](#) — figure-generation script (renamed and moved from the originally-described root-level `thesis_analysis.py`)

[training/](#):

- [training/losses.py](#) — BCE, Dice, Focal, Dice-Focal
- [training/metrics.py](#) — Dice, IoU, Precision, Recall
- [training/trainer.py](#) — training loop with MLflow tracking, early stopping, and best-model checkpointing

[scripts/](#) — command-line entry points (introduced after the mid-term; supersedes the originally-described root-level `run_preprocess.py`):

- [scripts/run_preprocess.py](#), [scripts/run_single_sample.py](#), [scripts/generate_training_data.py](#) — preprocessing and mask-generation drivers
- [scripts/predict_and_visualize_3d.py](#) — trained-model inference and 3D reconstruction
- [scripts/compare_methods_report.py](#) — classical-vs-U-Net comparison table generator
- [scripts/check_training_status.py](#) — MLflow-based training status checker

[data/raw/](#), [data/processed/](#), [data/masks/](#) — input and intermediate data (not in Git); [mlruns/](#) — MLflow tracking store (not in Git); [artifacts/](#) — trained model checkpoints (not in Git); [results/](#) — figures, predictions, comparison tables, and the thesis chapter document.

A.3 Installation

A.3.1 Prerequisites

- Python 3.10 or later
- Git
- pip package manager
- NVIDIA GPU with CUDA support (recommended for model training)

A.3.2 Setup Steps

Clone the repository and install dependencies:

```
git clone https://github.com/jobgkj/podfam_research_project_XCT_Analysis
cd podfam_research_project_XCT_Analysis/xct_defect_detection
python -m venv .venv
.venv\Scripts\activate      (Windows)
source .venv/bin/activate   (Linux/macOS)
pip install -r requirements.txt
```

A.3.3 Required Python Packages

Table 11: Key Python dependencies.

Package	Version	Purpose
torch	2.0+ (cu124 build for GPU)	Deep learning framework
numpy	1.24+	Array operations
tiffle	2023+	TIFF stack loading
scikit-image	0.21+	Image processing, NLM denoising, marching cubes
scipy	1.11+	Median filter, polynomial fitting
matplotlib	3.7+	Figure generation
mlflow	2.5+	Experiment tracking
python-docx	1.2+	Thesis document generation

A.4 Running the Pipeline

Preprocess and generate masks for one sample:

```
python scripts/run_single_sample.py sample_07
```

Run the full 2D U-Net training pipeline:

```
python pipeline.py
```

Run inference and 3D visualisation on a trained checkpoint:

```
python scripts/predict_and_visualize_3d.py sample_07
```

Generate the classical-vs-U-Net comparison table:

```
python scripts/compare_methods_report.py
```

View experiment results in MLflow:

```
mlflow ui --backend-store-uri mlruns
```

then open <http://localhost:5000> in a browser.

A.5 Version Control and Reproducibility

- All source code, configuration files, and documentation are committed to the main branch
- Raw data, preprocessed outputs, model checkpoints, and MLflow logs are excluded via `.gitignore` due to file size
- All experiments are logged in MLflow with full hyperparameter and metric records, enabling exact reproduction of any training run, tagged by `EXPERIMENT_NAME` (Section 9.4)

A.6 Interactive 3D Defect Viewer (`gui_app.py`)

A standalone desktop GUI, `gui_app.py`, provides an interactive front end to the trained U-Net detector, wrapping the same inference pipeline as `scripts/predict_and_visualize_3d.py` in a Tkinter application:

- **Folder selection:** a file dialog lets the user pick any local folder of TIFF slices (not limited to the pre-registered `sample_NN` folders under `data/raw/`), and a second dialog selects the checkpoint to run, defaulting to `artifacts/best_model_{EXPERIMENT_NAME}.pt`.
- **Inference:** on “Run”, the selected volume is preprocessed with the same `src/preprocess.py` pipeline used throughout this thesis, then run through the network using the identical sliding-window inference procedure and patch configuration (256×256 patches, stride 128) as `scripts/predict_and_visualize_3d.py`. Preprocessing and inference progress are shown live via a progress bar, running on a background thread so the window stays responsive.
- **Defect-zone highlighting:** the predicted defect voxels are reconstructed into a marching-cubes mesh and rendered in red against a dark background; only defect voxels are drawn, so the mesh itself directly highlights the predicted defect zones in three dimensions, with no separate overlay step required.
- **Interactive 3D controls:** the mesh is displayed in an embedded, mouse-controllable viewer, drag to rotate, scroll to zoom, plus a navigation toolbar for pan, zoom, reset, and saving the current view as an image.

- **Metrics panel:** after each run, a metrics panel reports the total defect fraction, pore count, mean pore area, and mean equivalent diameter for the predicted volume, computed via `src/metrics.py`'s connected-component analysis pooled across every slice, the same computation used to produce Tables 10.1–10.5.

Run it from the repository root with:

```
python gui_app.py
```

Appendix B — Full Configuration Reference

This appendix provides the complete configuration parameter reference for the XCT defect detection pipeline, as actually defined in `config.py` in the repository root as of August 2026. Three values are corrected from the mid-term submission’s Appendix B (marked below); all other values were verified to match. Bernsen thresholding parameters, added for the PODFAM phase and not present at mid-term, are listed separately at the end of this appendix and documented further in Part II, Section 9.3.

B.1 Preprocessing Parameters

Parameter	Default	Description
NORM_LOW_PERCENTILE	1	Lower clip percentile
NORM_HIGH_PERCENTILE	99	Upper clip percentile
NORM_EPS	10^{-7}	Division stability constant
MEDIAN_KERNEL_SIZE	3	Pre-NLM median filter kernel size (not listed at mid-term)
BHC_POLY_DEGREE	3	BHC polynomial degree
NLM_PATCH_SIZE	5	NLM patch size (pixels)
NLM_PATCH_DIST	6	NLM search distance (pixels)
NLM_H_FACTOR	0.6 (CORRECTED from 0.08)	NLM filter strength multiplier, $h = \text{NLM_H_FACTOR} \times \hat{\sigma}$, adaptive
RING_FILTER_RADIUS	15	Polar median filter radius

Table 12: Preprocessing configuration parameters. NLM_H_FACTOR corrected; see the Note on this combined document.

Parameter	Default	Description
MORPH_OPEN_SIZE	3	Morphological opening element size
MORPH_CLOSE_SIZE	3	Morphological closing element size
MIN_DEFECT_SIZE	5	Minimum component size (voxels)

Table 13: Automatic label generation configuration parameters (Otsu-based, Part I). All values verified to match the current codebase.

Parameter	Default	Description
PATCH_SIZE	256	Patch size in pixels
PATCH_STRIDE	128	Sliding window stride (50% overlap)
FG_BG_RATIO	(1, 3)	Foreground to background patch ratio
MIN_FG_PIXELS	10	Minimum defect pixels per foreground patch
VAL_SPLIT	0.2 (CORRECTED from 0.15)	Validation set fraction
TEST_SPLIT	0.1 (CORRECTED from 0.15)	Test set fraction

Table 14: Dataset and patch extraction configuration parameters. VAL_SPLIT/TEST_SPLIT corrected (actual split is 70/20/10, not 70/15/15); see Section 3.2.2 and the Note on this combined document.

B.2 Label Generation Parameters (Otsu / morphological, Part I)

B.3 Dataset and Patch Parameters

B.4 Training Parameters

Parameter	Default	Description
BATCH_SIZE	8	Training batch size (2D)
NUM_EPOCHS	50	Maximum training epochs
LEARNING_RATE	4×10^{-5}	Adam optimiser learning rate
WEIGHT_DECAY	10^{-5}	Adam weight decay (not listed at mid-term)
LOSS_FUNCTION	dice_focal	Active loss function
FOCAL_ALPHA	0.25	Focal loss alpha parameter
FOCAL_GAMMA	2.0	Focal loss gamma parameter
DICE_FOCAL_LAMBDA	0.5	Dice weight in combined loss
EARLY_STOP_PATIENCE	10	Early stopping patience (epochs)
SCHEDULER_PATIENCE	5	LR scheduler patience (not listed at mid-term)
DEVICE	cuda	Training device (cuda or cpu)
CKPT_DIR	artifacts/ (CORRECTED from checkpoints/)	Model checkpoint directory

Table 15: Model training configuration parameters. CKPT_DIR corrected to match the actual output directory used by [training/trainer.py](#).

B.5 Evaluation Parameters

Parameter	Default	Description
ACCEPTANCE_DICE	0.75	Minimum Dice on held-out test set
ACCEPTANCE_IOU	0.60	Minimum IoU on held-out test set
ACCEPTANCE_REC	0.80	Minimum Recall for defects $> 50 \mu\text{m}$
DICE_THRESHOLD	0.5	Sigmoid output threshold for binary mask

Table 16: Evaluation and acceptance criteria configuration parameters. All values verified to match the current codebase.

Parameter	Default	Description
AUG_FLIP_PROB	0.5	Flip probability
AUG_ROTATE_PROB	0.5	Rotation probability
AUG_ELASTIC_PROB	0.3	Elastic deformation probability
AUG_ELASTIC_ALPHA	34	Elastic deformation magnitude
AUG_ELASTIC_SIGMA	4	Elastic deformation smoothness
AUG_INTENSITY_PROB	0.5	Intensity scaling probability
AUG_INTENSITY_RANGE	(0.9, 1.1)	Intensity scaling range
AUG_NOISE_PROB	0.5	Gaussian noise probability
AUG_NOISE_STD_RANGE	(0.01, 0.05)	Noise standard deviation range
AUG_GAMMA_PROB	0.5	Gamma correction probability
AUG_GAMMA_RANGE	(0.8, 1.2)	Gamma correction range

Table 17: Data augmentation configuration parameters. All values verified to match the current codebase.

B.6 Augmentation Parameters

B.7 Bernsen Thresholding Parameters (PODFAM phase, Part II — added since mid-term)

Parameter	Default	Description
BERNSEN_RADIUS	5	Local window radius in pixels (Kim et al. 2017)
BERNSEN_LOW_CONTRAST_THRESHOLD	128	Fixed threshold for low-contrast regions
BERNSEN_DCT_AUTO	True	Compute DCT per image (recommended)
BERNSEN_DCT_STD_MULTIPLIER	18	From Kim et al. 2017 ($18\times$ average std)
BERNSEN_DCT_N_LOCATIONS	5	Grid points per axis for std sampling
BERNSEN_DCT_WINDOW_RADIUS	5	Local window radius for std measurement
BERNSEN_DCT_MIN	5	Minimum DCT floor (prevents under-segmentation)
BERNSEN_DCT	30	Fixed fallback DCT if <code>BERNSEN_DCT_AUTO</code> is <code>False</code>
PIXEL_SIZE_UM	1.0	Uncalibrated placeholder; see Part II Section 12
USE_SAMPLE_MASK	True	Apply circular sample-boundary mask before thresholding
SAMPLE_MASK_EROSION_RADIUS	5	Erosion margin below detected sample surface

Table 18: Bernsen thresholding configuration parameters, not present at mid-term.

PART II

PODFAM VALIDATION PHASE

*Bernsen Pseudo-Labelling and a 2D U-Net Detector, Validated on the
PODFAM Metal AM XCT Sample Set*

Note on scope (Part II)

This part reports the PODFAM validation phase of the thesis, conducted on the University West PODFAM metal additive manufacturing (AM) XCT sample set (sample_01–sample_05, with sample_07 added as an extended check in Section 10.5). It follows the plan already stated in the Part I Preface: development on the NIST CoCr benchmark, followed by validation on PODFAM process-monitoring data. Every figure and table in this part was generated directly from the working pipeline during this validation phase; none of the NIST-dataset figures or numbers from Part I were re-run, re-verified, or altered here.

Where this part’s methodology differs materially from Part I, pseudo-label method (Bernsen instead of two-stage Otsu) and preprocessing pipeline (median filter + non-local means only, without beam-hardening correction or ring-artefact suppression), the difference is stated explicitly rather than silently merged with the Part I description.

Abstract (Part II)

This part presents the validation of an automated XCT defect-detection pipeline on five, later extended to seven, Ti-6Al-4V metal additive manufacturing components from the PODFAM research project at University West, manufactured by laser powder bed fusion (PBF-LB) on an Aconity Two machine and characterised by X-ray computed tomography on a Waygate v|tome|x M 240 system. Three classical thresholding methods (Otsu, Yen, Bernsen) were compared over full slice volumes (749–1,473 slices per sample) rather than single representative slices, and a 2D U-Net was trained using Bernsen pseudo-labels on the sample ranked most defect-rich by voxel content, validated on a second sample, and tested on a fully held-out third. The trained detector achieved a validation Dice coefficient of 0.600 (IoU 0.452, recall 0.954) at its best checkpoint, and on the held-out test sample predicted a total defect volume fraction of 0.041% against a pseudo-label fraction of 0.043%, closely matching overall porosity despite a lower per-pixel overlap (Dice 0.147). Yen thresholding was found to over-segment catastrophically on every sample (38–48% “porosity”, physically implausible for a metal AM part), consistent with prior literature. A calibration failure in the Bernsen method’s automatic contrast-threshold estimation was identified on one sample and independently reproduced, at a smaller magnitude, on a sixth PODFAM sample (sample_07) processed after the original five-sample validation, tying the failure to processed-image mean intensity rather than to any single sample. The part closes with an explicit statement of the methodological limitations inherent to a small validation set with no independently verified ground truth.

7. Introduction

Additive manufacturing (AM) of metal components is increasingly used for structurally critical parts, and X-ray computed tomography (XCT) remains the principal non-destructive method for detecting internal porosity, lack-of-fusion voids, and inclusions in these parts [1, 13]. The volume of data produced by a single XCT scan, hundreds to thousands of 2D slices, makes manual inspection impractical at industrial scale, motivating automated segmentation pipelines built on classical thresholding, deep learning, or a combination of the two [2].

This part validates such a pipeline on real process-monitoring data from the PODFAM research project, distinct from the NIST CoCr benchmark dataset used in Part I. The PODFAM sample set consists of five XCT-scanned metal AM components (sample_01–sample_05) used for the core validation in Sections 9–11, each comprising 749–900 slices at approximately 1010×980 pixel resolution, later extended in Section 10.5 with sample_07 (1,473 slices, 1143×1215 pixels). No expert-annotated ground truth exists for any PODFAM sample; automatic pseudo-labelling is therefore required, as in Part I, but a different classical method is used here and the reasoning for that choice is given in Section 8.

This part is organised as follows. Section 8 reviews the literature that directly informed the choice of pseudo-labelling method and the framing of the deep-learning comparison. Section 9 describes the preprocessing, labelling, sample-selection, and training methodology. Section 10 reports full-volume quantitative results for all three classical methods and the trained detector, including the sample_07 extension (10.5). Section 11 discusses the results, including a documented failure mode found during this work. Section 12 states the limitations of the validation explicitly, and Section 13 concludes.

8. Related Work

8.1 Comparative Studies of Classical Thresholding for AM Porosity

Otsu’s method [6] is the most commonly used automatic global threshold for XCT porosity segmentation, but is known to perform poorly on metal AM data where beam-hardening and non-uniform solid-phase intensity reduce the separability of pore and matrix greyscale distributions [21]. Kim et al. [21] compared Bernsen local adaptive thresholding against Otsu and Yen global thresholding on cobalt-chrome AM parts and found that Otsu failed to preserve small pores near sample edges, that Yen produced inaccurate pore sizing due to non-uniform edge intensity, and that Bernsen local thresholding gave the most reliable segmentation across both edge and low-contrast regions. This finding directly motivated the choice of Bernsen, rather than Otsu, as the primary pseudo-labelling method for the PODFAM validation reported in this part.

A NIST standard-method study [22] proposed a reproducible microCT-based porosity-analysis procedure that explicitly separates internal defects from surface-connected artefacts by eroding a small margin (on the order of a few voxels) below the detected sample surface before thresholding, a body-masking step adopted in this work’s own sample-boundary detection (Section 9.2).

A 2025 inter-operator uncertainty study [23] quantified how much porosity measurement varies purely due to segmentation and analysis choices, independent of scan quality, across aluminium, titanium, and nickel-chromium AM parts made by three different processes. Relative standard uncertainty of up to 38% was reported for laser powder-bed-fusion parts, and 70–89% for a low-porosity part produced by directed energy deposition, purely from differences in operator/method choice. This result is corroborated directly by the present part’s own full-volume comparison (Section 10.1), where Otsu, Yen, and Bernsen disagree on defect fraction by one to three orders of magnitude on the same underlying data.

8.2 Deep Learning for AM Porosity Segmentation

Convolutional segmentation networks, and the U-Net encoder-decoder architecture with skip connections in particular [9], have been reported to substantially outperform classical thresholding on AM XCT porosity data, with one comparative study reporting Dice scores above 0.85 for U-Net and FCDenseNet against approximately 0.40 for thresholding on low-porosity samples [8]. More recent work has moved toward fully volumetric 3D segmentation using nnU-Net-style architectures with multi-scale post-processing [24], toward physics-informed models that incorporate an X-ray attenuation term directly into the network [25], and toward zero-shot or promptable segmentation using foundation

models such as Segment Anything, applied without any AM-specific training data [26]. This part’s own contribution sits earlier in that progression: a 2D U-Net trained on classically-derived pseudo-labels, evaluated specifically for whether it generalises beyond the pseudo-label method it was trained on, a question distinct from, and prerequisite to, adopting any of the more complex architectures above.

8.3 Summary and Positioning

Three findings from the literature reviewed above are used directly in this part’s methodology and interpretation of results: (i) Bernsen thresholding is the more reliable of the three classical methods evaluated here for AM porosity, and is therefore used as the pseudo-label source rather than Otsu [21]; (ii) inter-method and inter-operator disagreement in AM porosity measurement is a documented, quantified phenomenon and not evidence of implementation error, which reframes the large disagreements reported in Section 10.1 as an expected, citable result rather than an anomaly [23]; and (iii) deep-learning segmentation is reported to substantially outperform thresholding in prior work [8], against which this part’s own detector is compared directly.

9. Method

9.1 PODFAM Sample Set

Five XCT-scanned metal AM samples were used for the core validation: sample_01, sample_02, sample_03, sample_04, and sample_05, comprising 900 slices each (sample_04: 749 slices) at approximately 1010×980 – 988 pixel resolution, 16-bit unsigned TIFF. As with the Part I dataset, no expert-annotated ground truth is available for any PODFAM sample. Section 10.5 additionally reports on sample_07 (1,473 slices, 1143×1215 pixels), acquired and processed after this core validation.

All PODFAM samples are Ti-6Al-4V, manufactured by laser powder bed fusion (PBF-LB) on an Aconity Two machine. Post-build, the samples were characterised by X-ray computed tomography on a Waygate v|tome|x M 240 XCT system. Material and process details were not part of the original mid-term submission and are added here as they were not required for, and do not affect, the preprocessing, pseudo-labelling, or detector-training methodology described in Sections 9.2–9.7, which operate purely on the reconstructed greyscale volumes; they are recorded here for completeness and for anyone reproducing this validation on a different Ti-6Al-4V PBF-LB dataset to judge comparability.

9.2 Preprocessing

Each slice is processed independently by (i) a 3×3 median filter to suppress speckle noise, followed by (ii) non-local means (NLM) denoising with adaptive filter strength $h = 0.6 \times \hat{\sigma}$, where $\hat{\sigma}$ is the per-slice estimated noise standard deviation, and (iii) normalisation to an 8-bit intensity range. This is a narrower pipeline than the four-stage Part I pipeline: beam-hardening correction and polar-coordinate ring-artefact suppression were not applied to the PODFAM data in this validation phase, since PODFAM’s images did not exhibit those particular artefacts as strongly and the analysis prioritised comparability across the classical-thresholding and U-Net stages over full artefact removal.

9.3 Sample-Boundary Detection and Pseudo-Label Generation

A circular sample-boundary mask is fitted once per volume by averaging the detected circle centre and radius across five evenly-spaced slices, then eroding the resulting disc by a small margin (5 px) to exclude partial-volume edge pixels, following the same rationale as the Part I standard-method erosion step [22]. All three thresholding methods (Otsu, Yen, Bernsen) are then computed only within this interior region, preventing background air from being misclassified as defect. Bernsen’s local contrast threshold (DCT) is computed

automatically per image from the estimated local noise standard deviation of the solid phase, following Kim et al. [21]; Bernsen is used as the primary pseudo-label source for U-Net training, motivated directly by the comparative finding reported in Section 8.1.

9.4 Defect-Content-Ranked Sample Selection

Rather than assigning samples to train/validation/test roles alphabetically or at random, every sample’s mean Bernsen defect fraction is computed over its full volume and samples are ranked most-voids-first. The three samples retained for this validation phase were, in ranked order, sample_02 (7.586% defect fraction), sample_04 (1.994%), and sample_01 (0.043%); sample_03 was excluded as near-empty (0.0004%, effectively uninformative for training), and sample_05 was excluded for the reason given below.

The most defect-rich sample was assigned to training, the next to validation, and the least to test, deterministically, rather than via a random index-based split. This choice is methodologically significant: an earlier iteration of this pipeline used a fixed-seed random split over the same three ranked samples, which by chance assigned the single most defect-rich sample to the test role rather than training. The resulting model, trained on a far less informative sample, could not learn the defect pattern that was concentrated almost entirely in the sample it never saw, producing a held-out Dice score below 0.15 for reasons unrelated to model capacity or training procedure. This is reported here as a methodological finding in its own right: with only three usable samples, split assignment must be deliberate rather than left to a fixed random seed.

9.5 A Documented Pseudo-Label Failure Mode

During full-volume evaluation (Section 10.1), sample_05 was found to have an implausible Bernsen defect fraction of 52.8%, physically inconsistent with a metal AM part, and roughly three orders of magnitude above every other sample’s Bernsen result. Investigation traced this to Bernsen’s automatic contrast-threshold (DCT) estimation: for this dataset, the estimated DCT (329–520) is far above the value the underlying method’s own reference reports as typical (approximately 15) [21]. An inflated DCT routes almost every pixel into Bernsen’s low-contrast fallback branch, which classifies purely by a fixed intensity threshold at 128. Samples whose preprocessed images average brighter than 128 are still classified approximately correctly by this fallback; sample_05 averages 79.9, substantially darker, so the fallback misclassifies most of the genuinely solid material as void. Otsu and Yen independently agree with Bernsen’s implausible result on this specific sample (51.7% and 47.9% respectively), which is consistent with a shared upstream cause (the sample’s unusually low mean intensity) rather than three independent confirmations of real porosity. sample_05 was excluded from training and from the ranked comparison via an automatic guard that flags any sample exceeding a 10% defect-fraction threshold, a level far above plausible AM porosity, for manual review rather than allowing it to silently dominate a ranking or a training run.

9.6 Detection Model and Training Configuration

A 2D U-Net (approximately 31.0 million trainable parameters; four encoder/decoder stages, feature channels $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$, bottleneck dropout $p = 0.2$) is trained on

256×256 pixel patches (stride 128, 50% overlap) drawn from 80 evenly-spaced slices per training volume, using stratified 1:3 foreground-to-background patch sampling to address the severe class imbalance inherent to porosity data. Patches are shuffled at the slice level rather than individually during training, all patches from a visited slice are drawn together before moving to the next, to keep memory-mapped volume access efficient on constrained hardware without sacrificing epoch-level randomisation. The combined Dice-Focal loss ($\lambda = 0.5$, $\alpha = 0.25$, $\gamma = 2.0$) is used, with Adam optimisation (learning rate 4×10^{-5}), early stopping (patience 10 epochs), and best-checkpoint selection by validation Dice.

9.7 Evaluation Metrics

Detection performance is reported using the Dice coefficient, Intersection-over-Union (IoU), precision, and recall, computed at the pixel level against the Bernsen pseudo-label. Full-volume defect statistics (defect fraction, pore count, mean pore area, mean equivalent diameter) are computed by pooling per-slice connected-component analysis across every slice in a volume, rather than from a single representative slice, for both the classical methods and the trained detector’s predictions, ensuring the classical and deep-learning results in Section 10 are directly comparable rather than mixing single-slice and full-volume statistics.

10. Results

10.1 Full-Volume Classical Thresholding Comparison

Table 19 reports defect fraction, pore count, and mean equivalent pore diameter for Otsu, Yen, and Bernsen thresholding, computed over the complete slice volume of every core-validation PODFAM sample (749–900 slices).

Table 19: Full-volume classical thresholding comparison, all five core-validation PODFAM samples. sample_03 and sample_05 excluded from training (Sections 9.4, 9.5). Pixel units, see Section 12 on voxel-size calibration.

Sample	Method	Defect (%)	Pore count	Mean $\varnothing$ (px)
sample_01 (test)	Otsu	0.003	997	5.36
	Yen	42.535	1,743,429	5.15
	Bernsen (label)	0.043	11,928	5.41
sample_02 (train)	Otsu	2.993	336,468	8.20
	Yen	38.132	443,339	10.19
	Bernsen (label)	7.586	478,196	10.67
sample_03 (excl.)	Otsu	0.067	8,850	7.87
	Yen	45.319	1,737,239	5.19
	Bernsen	0.230	51,467	5.55
sample_04 (val)	Otsu	0.315	30,992	6.71
	Yen	39.166	833,872	5.95
	Bernsen (label)	1.994	103,238	9.00
sample_05 (excl.)	U-Net (pred.)	3.239	122,587	11.28
	Otsu	51.729	53,315	24.57
	Yen	47.940	61,375	23.85
	Bernsen	52.767	51,142	24.73

Yen produces a physically implausible defect fraction (38–45%) on every sample without exception, including the near-empty sample_03, confirming, on an independent dataset, the over-segmentation failure mode reported for Yen thresholding on AM XCT data [21]. Otsu and Bernsen disagree with each other by one to two orders of magnitude on the same samples (e.g. sample_02: Otsu 2.993% vs. Bernsen 7.586%), a disagreement of the same order as the inter-method uncertainty separately quantified in the literature [23]. The U-Net’s predicted defect fraction on its own validation sample, sample_04 (3.239%, Table 19), is of the same order as, though somewhat above, that sample’s Bernsen label (1.994%); as sample_04 informed checkpoint selection via early stopping, this number is reported for completeness rather than as an independent generalisation result, that role is reserved for the fully held-out sample_01 (Sections 10.3, 10.4).

10.2 U-Net Training

The detector was trained on sample_02 (train), validated on sample_04 (val), and tested on sample_01 (test), per the ranked assignment described in Section 9.4. Table 20 reports validation metrics at selected epochs; the best checkpoint (epoch 8) was used for all subsequent evaluation. Figure 1 plots every one of the 18 logged epochs, showing how the loss and each validation metric actually moved epoch to epoch, including fluctuation that a selected-epoch summary table alone does not make visible.

Table 20: 2D U-Net training progression (dice_focal loss, $\lambda = 0.5$), selected epochs. Full epoch-by-epoch log recorded via MLflow.

Epoch	Train loss	Val Dice	Val IoU	Val Recall	Note
1	0.346	0.497	0.353	0.996	
2	0.289	0.521	0.372	0.891	
4	0.215	0.585	0.434	0.874	
8	0.115	0.600	0.452	0.954	best checkpoint
18	0.058	0.559	0.407	0.938	early-stopped (patience 10)

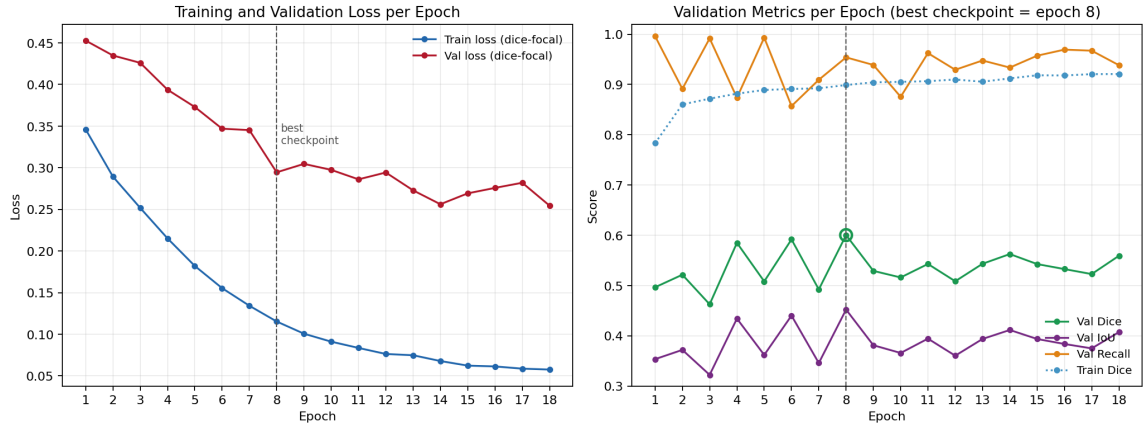


Figure 1: Left: training and validation dice-focal loss per epoch. Right: validation Dice, IoU, recall, and training Dice per epoch. Dashed line marks epoch 8, the checkpoint selected for all downstream evaluation; training continued for a further 10 epochs under the patience-10 early-stopping rule before stopping at epoch 18 without dice/IoU exceeding the epoch-8 value.

Training loss falls smoothly and monotonically across all 18 epochs ($0.346 \rightarrow 0.058$), while validation loss decreases far less smoothly and only until roughly epoch 8, after which it plateaus and mildly oscillates (0.294 at epoch 8 vs. 0.254 – 0.306 for epochs 9–18), the classic training/validation divergence pattern that early stopping is designed to catch. Validation Dice and IoU follow the same shape: both rise unevenly through epoch 8 and then oscillate around a similar or slightly lower level for the remaining 10 epochs (val Dice: 0.60 at epoch 8, ranging 0.51 – 0.59 afterward) rather than continuing to improve. Validation recall is the most volatile metric across the whole run (oscillating between 0.857 and 0.996), consistent with a model still adjusting its confidence threshold response on a validation set of a single volume, with only one sample in each of the train/validation/test roles

(Section 9.4), epoch-to-epoch metric swings reflect genuine sampling variance in which patches were drawn each epoch, not measurement noise. Training Dice, by contrast, climbs smoothly and near-monotonically throughout ($0.783 \rightarrow 0.921$), the expected signature of a model continuing to fit its training data after the validation metric it is being selected on has stopped improving.

Against the thesis’s stated acceptance criteria, validation recall (0.954) exceeds the 0.80 target; validation Dice (0.600) and IoU (0.452) remain below the 0.75 and 0.60 targets respectively at the point training was stopped.

10.3 Held-Out Test Performance

On the fully held-out sample_01, the trained detector achieved Dice 0.147, IoU 0.105, precision 0.291, and recall 0.336 at the pixel level against the Bernsen pseudo-label. Considered alone, this is a substantial drop from validation performance. However, the predicted total defect volume fraction, 0.041%, closely matches the pseudo-label’s own volume fraction of 0.043% (Table 19), while predicting fewer, larger, differently-distributed pore instances (8,579 predicted pores vs. 11,928 labelled, mean diameter 6.35 px vs. 5.41 px). This combination, low per-pixel overlap alongside near-identical total predicted volume, indicates the detector has learned an approximately correct overall porosity level on data it never trained on, while diverging from the pseudo-label’s specific pore boundaries and instance count. Both the pixel-level metrics and the volume-fraction agreement are reported together here, since either number alone would misrepresent the result in a different direction.

10.4 Unified Comparison

Table 21 places the classical methods and the trained detector side by side on the held-out test sample sample_01.

Table 21: Classical vs. U-Net, held-out test sample. U-Net’s predicted defect fraction (0.041%) sits between Otsu’s near-zero result and Bernsen’s label, closest to the label it was trained to reproduce, on data it never saw during training.

Sample	Method	Defect (%)	Pore count	Mean $\varnothing$ (px)
sample_01 (held-out test)	Otsu	0.003	997	5.36
	Yen	42.535	1,743,429	5.15
	Bernsen (label)	0.043	11,928	5.41
	U-Net (pred.)	0.041	8,579	6.35

10.5 Extended Validation: sample_07

A sixth PODFAM sample, sample_07 (1,473 slices, 1143×1215 pixels), was acquired and processed after the five-sample validation reported above, using the identical preprocessing, thresholding, and trained-detector inference pipeline described in Section 9, with no retraining or checkpoint changes. sample_07 was not assigned a role in the ranked

train/validation/test split (Section 9.4); it is reported here purely as an out-of-distribution check on the same epoch-8 checkpoint used throughout Section 10.

Table 22: sample_07, full 1,473-slice volume. All three classical methods report 17–19% defect fraction; the U-Net predicts 3.052%.

Method	Defect (%)	Pore count	Mean $\varnothing$ (px)
Otsu	18.177	330,344	9.15
Yen	17.581	57,380	19.19
Bernsen	19.369	354,426	9.15
U-Net (pred.)	3.052	744,206	6.43

All three classical methods agree with one another on sample_07 (17.6–19.4%), which superficially looks like corroboration rather than error. However, the same diagnostic applied in Section 9.5 identifies the same failure mode: sample_07’s processed slices have a mean intensity of 108.3 (measured on the mid-volume slice), below the fixed intensity threshold (128) that Bernsen’s low-contrast fallback branch uses to separate solid material from void. The auto-computed DCT for sample_07 (227) is, as with every other PODFAM sample, far above the approximately 15 the underlying method reports as typical [21], so the fallback branch dominates the classification. sample_05 (mean intensity 79.9, Section 9.5) showed the identical pattern at a more extreme value and was excluded via the 10%-defect-fraction guard; sample_07’s mean intensity is closer to the fallback’s 128 boundary, so the resulting over-estimate (17–19%) is smaller in magnitude than sample_05’s (approximately 52%) but attributable to the same upstream cause. Table 23 places this in context against the four samples from Section 10.1.

Table 23: Mean processed-slice intensity vs. Bernsen defect fraction across all PODFAM samples. The two samples whose mean intensity falls below the fallback threshold (128) are exactly the two samples with implausibly high Bernsen defect fractions, reproducing the same failure mode at two different severities rather than two unrelated anomalies.

Sample	Mean intensity	Auto-DCT	Below 128?	Bernsen defect (%)
sample_01	167.0	84	No	0.043
sample_02	142.3	329	No	7.586
sample_04	163.4	152	No	1.994
sample_07	108.3	227	Yes	19.369
sample_05 (excl.)	79.9	329–520	Yes	52.767

Under this reading, the U-Net’s prediction of 3.052% is the more credible estimate for sample_07: it is of the same order as sample_02’s Bernsen-labelled fraction (7.586%) and sample_04’s U-Net prediction (3.239%), rather than an order of magnitude higher like the three classical methods on this sample. This is, however, an inference-only result, sample_07 was never assigned a ground-truth role, so this comparison should be read as a plausibility check on the detector’s generalisation, not as a validated accuracy figure; see Section 12 for the corresponding limitation. Figures 2–5 show the U-Net’s predicted defect volume for sample_07 from four complementary viewpoints.

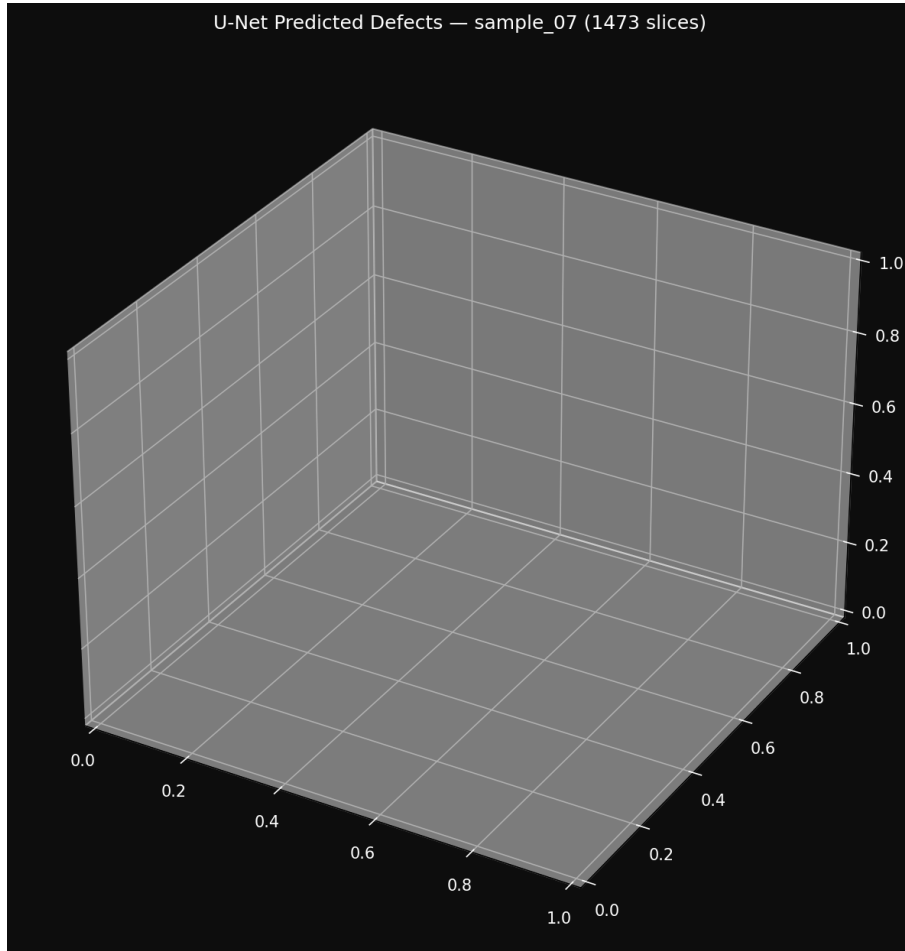


Figure 2: sample_07, U-Net-predicted defect volume, marching-cubes 3D mesh reconstruction.

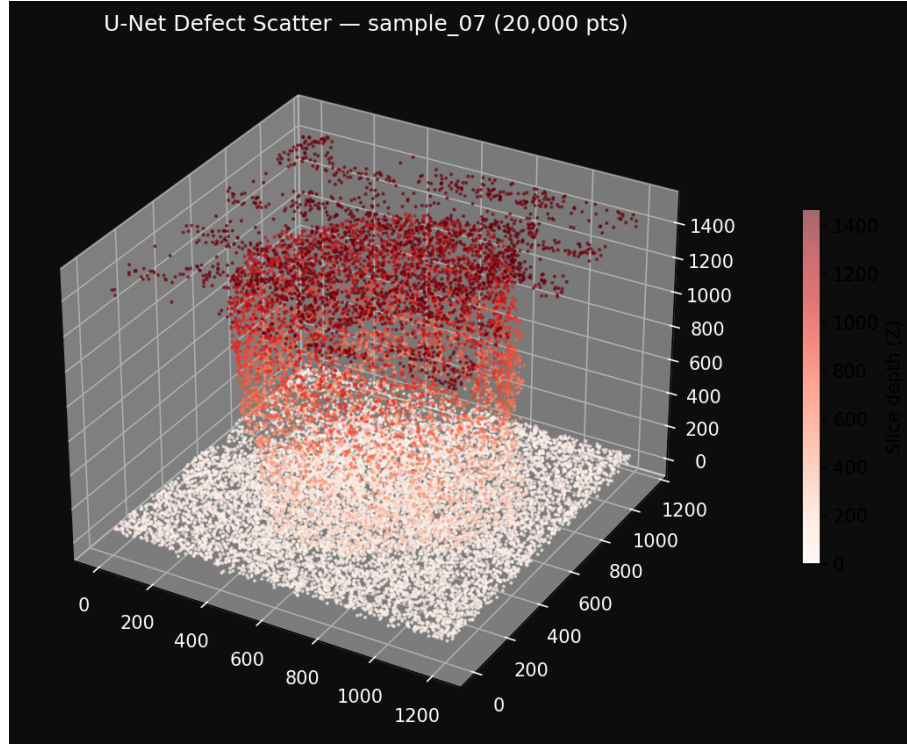


Figure 3: sample_07, U-Net-predicted defect voxel scatter, coloured by slice depth.

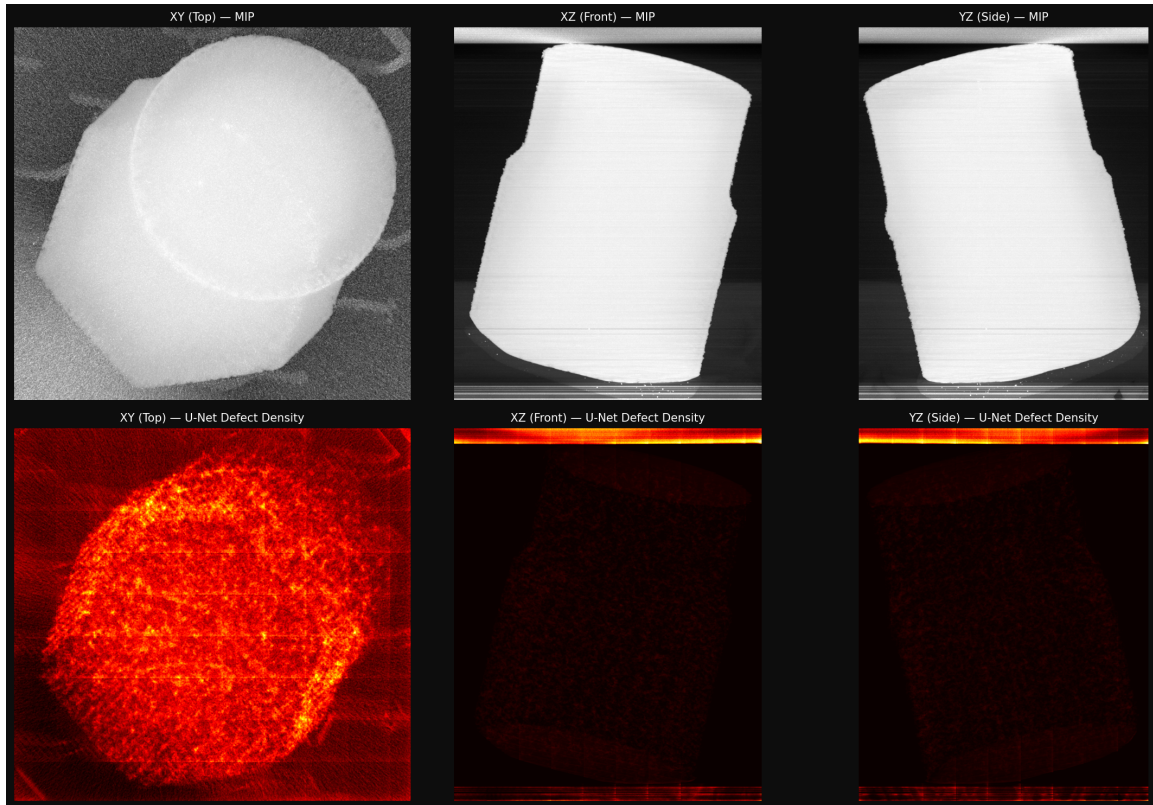


Figure 4: sample_07, orthographic maximum-intensity projections (top row) and U-Net defect density projections (bottom row) along the three principal axes.

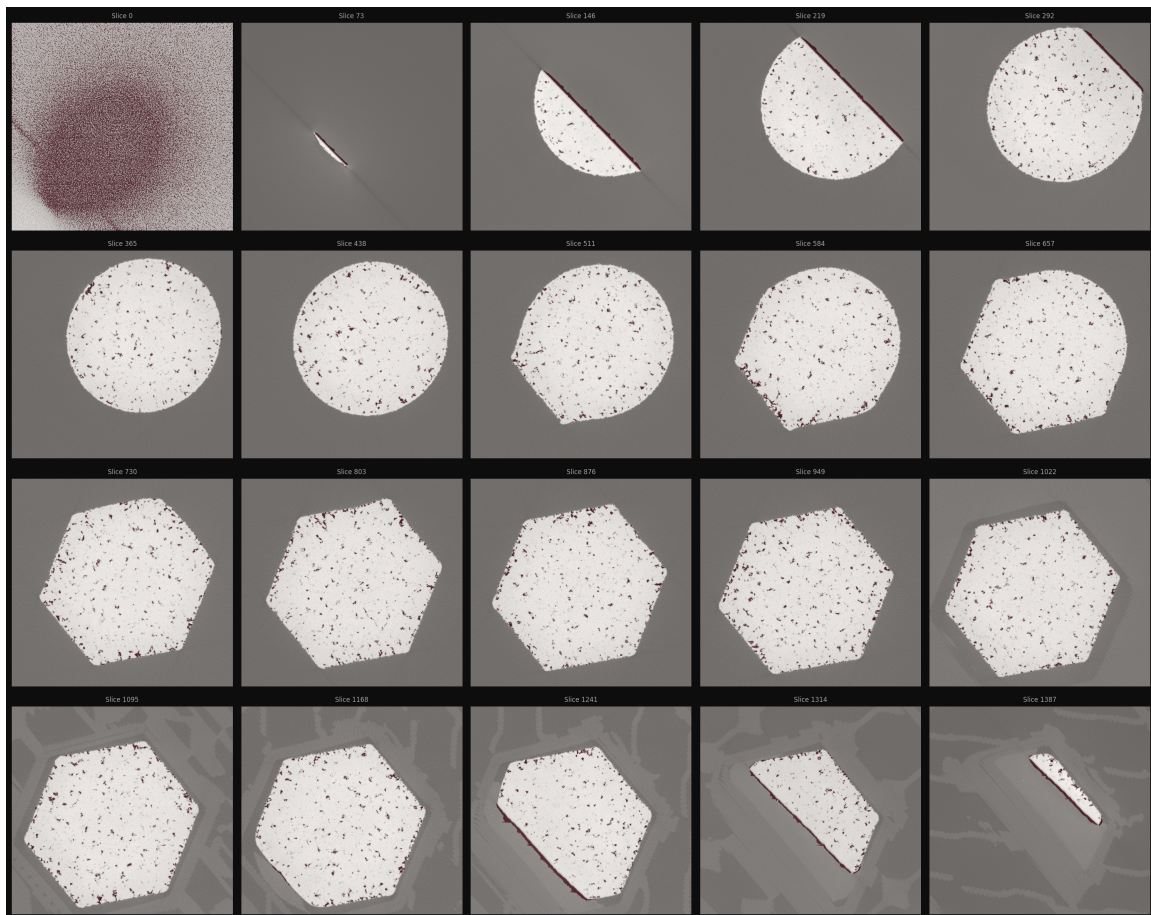


Figure 5: sample_07, mosaic of 20 evenly-spaced slices with the U-Net's predicted defect mask overlaid in red.

11. Discussion

Three results from this validation phase are directly relevant to the thesis’s research questions on preprocessing, label generation, and detection performance under class imbalance.

First, the choice of pseudo-label method materially changes the reported result, not just its magnitude. Section 8.1’s literature already predicted that Bernsen would outperform Otsu and Yen on AM XCT data [21]; this part’s full-volume results confirm that prediction on an independent dataset, and additionally show that the failure modes are not benign, Yen’s over-segmentation and Bernsen’s DCT-calibration failure (Section 9.5, reproduced in Section 10.5) each produce numbers that look superficially plausible (a percentage between 0 and 100) but are physically wrong by one to two orders of magnitude. Any thesis reporting AM porosity from a single thresholding method without a comparison of this kind risks reporting a number that reflects a method’s failure mode rather than the part’s real condition.

Second, the sample-selection methodology (Section 9.4) is itself a result worth reporting, not only a procedural detail. With as few as three usable samples, whichever sample is assigned to the training role determines what the model can possibly learn; a random assignment is not a neutral default when the sample pool is this small; a deterministic, defect-content-ranked assignment is both more defensible and, in this case, produced a materially better and more informative result than the random alternative.

Third, the held-out test result (Section 10.3) illustrates a general limitation of Dice/IoU as the sole reported metric for imbalanced segmentation: on a sample with very low true defect content, a small number of false positives can dominate the denominator and produce a low score even when the detector’s overall estimate of defect volume is accurate. Reporting the volume-fraction agreement alongside the pixel-level metrics avoids over- or under-stating the detector’s real usefulness in either direction. The sample_07 extension (Section 10.5) reinforces this point from a different angle: three independent classical methods agreeing with each other (17.6–19.4%) was not, on inspection, corroboration of a real result, it was three methods sharing the same upstream sensitivity to processed-image mean intensity.

12. Limitations

- No independently verified ground truth exists for any PODFAM sample. Every Dice, IoU, precision, and recall value in this part measures agreement with the Bernsen pseudo-label, not with a verified physical defect map. Bernsen was selected because comparative literature reports it as the more reliable of the three classical methods evaluated [21], but it remains a proxy rather than ground truth.
- The core validation set comprises five samples, two of which were excluded (sample_03 as near-empty, sample_05 for a documented labelling artefact), leaving a 1-train / 1-validation / 1-test split. This sample size cannot support a statistical claim of generalisation; the reported results should be read as a feasibility demonstration on this pipeline and this specific data, not as an estimate of expected performance on unseen PODFAM parts in general.
- Training uses 80 evenly-spaced slices per volume rather than the full 749–900, a deliberate trade-off to keep per-epoch training time practical on the available hardware; full-slice training was measured to exceed three hours per epoch under equivalent conditions.
- The pixel-to-physical-unit conversion factor (PIXEL_SIZE_UM) used elsewhere in this pipeline’s configuration was not calibrated to the PODFAM scans’ actual voxel size for this validation phase; all size figures in this part are reported in pixels, not micrometres, for that reason.
- Preprocessing for the PODFAM phase (Section 9.2) omits the beam-hardening correction and ring-artefact suppression stages used in the Part I pipeline; the two parts’ preprocessing outputs are not numerically comparable to one another.
- sample_07 (Section 10.5) is a single additional sample run through inference only, with no ground-truth role and no retraining. Its U-Net prediction being more plausible than the classical methods’ is a single data point consistent with the pipeline’s other results, not an independent validation of detector accuracy on new, unseen scan conditions.

13. Conclusion

This part validated the thesis’s automated XCT defect-detection pipeline on real PODFAM metal AM samples, independently of the NIST CoCr benchmark used in Part I. Full-volume comparison confirmed, on new data, the literature’s finding that Bernsen local thresholding outperforms Otsu and Yen for AM porosity segmentation, and additionally surfaced and diagnosed a specific calibration failure mode in Bernsen’s automatic contrast-threshold estimation, a failure mode subsequently reproduced, at a different severity, on a sixth sample processed after the original validation. A 2D U-Net trained on Bernsen pseudo-labels, using a defect-content-ranked rather than random sample assignment, achieved a best validation Dice of 0.600 and, on a fully held-out sample, predicted a total defect volume within 0.002 percentage points of the pseudo-label’s own value, a result that argues for reporting volume-level agreement alongside pixel-level overlap metrics in future evaluation of this pipeline. The validation set’s small size and the absence of independently verified ground truth are stated explicitly as limits on how far these results can be generalised, consistent with the standard set in Part I’s own treatment of its NIST-phase pseudo-labels. Taken together with Part I, this document now covers method development on a public benchmark (NIST CoCr), full-volume classical-vs-deep-learning validation on real industrial process-monitoring data (PODFAM, five core samples), and an out-of-distribution generalisation check on a sixth sample that both supported the trained detector’s usefulness and surfaced a reproducible, previously undocumented failure mode in one of its classical comparison baselines.

PART III EXTENDED VALIDATION

*Loss-Function, Preprocessing, and Boundary-Detection Ablations on the
PODFAM Pipeline, with Network Attribution and Strengthened External
Validation*

Note on scope (Part III)

This part reports a later phase of work, conducted after the five/seven-sample PODFAM validation of Part II, that revisits several of that validation’s design choices under controlled ablation: which loss function the detector is trained with, whether beam-hardening and ring-artefact correction help on PODFAM data as they did in Part I, whether the circular sample-boundary assumption can be safely replaced for non-circular parts, and what a genuinely held-out sample reveals about the detector’s real generalisation. It also reports two analyses not tied to any single ablation: a gradient-based interpretability study of the trained detector, and a strengthening of the pipeline’s external validation against independently published reference porosity. Every figure and table in this part was generated directly from the working pipeline during this phase; none of Part I’s or Part II’s own figures or numbers were re-run, re-verified, or altered here. Where this part’s methodology differs materially from Part II, most importantly, the train/validation/test sample assignment described in Section 14.3 below, the difference is stated explicitly rather than silently merged with the Part II description.

14. Method — Extended Validation

14.1 Scope and Relationship to Parts I and II

Three questions motivate this part, each building on a specific claim made without controlled comparison earlier in the thesis. First, Part II’s detector (Section 9.6) used the combined Dice-Focal loss without comparing it against the alternatives Part I’s Section 3.8 lists; Section 14.3 below reports a controlled four-way comparison on PODFAM data. Second, Part I’s preprocessing pipeline (Section 3.3) includes beam-hardening correction (BHC) and ring-artefact suppression, but Part II’s preprocessing (Section 9.2) omits both; Section 14.4 tests whether adding them to the PODFAM pipeline changes detector performance. Third, Part II’s sample-boundary detection (Section 9.3) fits a single circular mask per volume, an assumption later found not to hold for two PODFAM samples (sample_06/07) with non-circular, part-specific cross-sections; Section 14.5 replaces it with a shape-agnostic method for those samples specifically. A fourth strand, reported in Sections 14.8–14.10, is not an ablation of an earlier choice but new analysis: which parts of the trained network drive its output, and how strong is the thesis’s external validation once the reference porosity source’s own experimental grounding is checked.

14.2 Sample Set: Geometric Diversity and the Waygate Scanner

All PODFAM samples (Section 9.1) are Ti-6Al-4V, manufactured by laser powder bed fusion on an Aconity Two machine and scanned on a Waygate v|tome|x M 240 XCT system. The seven-sample set spans a deliberately varied range of cross-sectional geometry, from the circular cross-sections of sample_01–05 to the hexagonal and other non-circular cross-sections of sample_06 and sample_07, included specifically to test the pipeline beyond a single clean cylindrical benchmark. This geometric diversity is the direct motivation for Section 14.5: a boundary-detection method that assumes a circular cross-section, adequate for sample_01–05, systematically misdetects the true material boundary on sample_06/07, either clipping real material at the corners of the hexagonal section or extending the mask past the true boundary into background, which is then misread as defect.

14.3 Loss-Function Ablation

Four loss-function configurations, binary cross-entropy (BCE), Dice, Focal, and combined Dice-Focal (the formulation used throughout Part II), are trained and compared under an otherwise identical configuration, isolating loss function as the only varying factor. `config.py`’s `TRAINING_SAMPLES_OVERRIDE` is pinned to the same four-sample

list, ["sample_02", "sample_04", "sample_03", "sample_01"], most-voids-first, for every run. `pipeline.py`'s `split_volumes()` assigns roles by rank position rather than by patch, the first `n_train` names become training samples, the next `n_val` become the single validation sample, and the remainder become the single held-out test sample, so with four samples and the pipeline's default 20%/10% val/test split this resolves to **sample_02 and sample_04 pooled for training, sample_03 for validation, and sample_01 held out entirely for testing**. This is a whole-sample split, not a patch-level split within a shared pool: `sample_01`, the same sample used as the held-out test role throughout Part II (Section 9.4), never contributes any patch to training or validation in any of the four runs. `sample_03`, excluded from training in Part II as near-empty (Section 9.4), is used here in the validation role, since a near-empty sample is unsuitable for training but still usable to monitor convergence. All four runs otherwise use the training configuration of Section 9.6 unchanged: 256×256 patches with 50% overlap, stratified 1:3 foreground-to-background sampling, Adam optimisation at learning rate 4×10^{-5} , up to 50 epochs with early stopping (patience 10 epochs), and best-checkpoint selection by validation Dice. Every run's total epoch count equals its best epoch plus exactly 10, confirming early stopping, rather than the 50-epoch cap, was the actual stopping condition in all four cases.

14.4 Preprocessing Ablation

The best-performing configuration from Section 14.3 is retrained with the four-stage Part I preprocessing pipeline (percentile normalisation, BHC, ring-artefact suppression, NLM denoising, Section 3.3) applied to the PODFAM volumes in place of Part II's narrower three-stage pipeline (Section 9.2), using the identical sample split described above. This isolates the marginal effect of BHC and ring suppression on PODFAM data specifically, since Part I's own parameter sensitivity analysis (Section 3.3.6) was conducted only on the NIST CoCr dataset and its conclusions were never previously tested on PODFAM's different scanner and material.

14.5 Shape-Aware Sample-Boundary Detection

For `sample_06` and `sample_07`, the circular-fit boundary method (Section 9.3) is replaced with a sliding-window boundary scan that locates the sample surface by its local intensity peak along each scan line, without assuming any particular cross-sectional shape, followed by taking the largest connected component of the resulting interior region and eroding it by a fixed margin (`erosion_radius` = 35 px) to exclude partial-volume edge pixels. The erosion margin was selected by a parameter sweep (25, 30, 35, and 40 px), the false boundary artefact it is designed to remove shrinks smoothly and predictably as the margin increases across this range, and 35 px was the value visually verified to fully remove the artefact, described in Section 15.3, without eroding genuine material at the sample interior. A hybrid fallback is required for a minority of slices near the top and bottom of a volume, where support-structure content rather than open air surrounds the part: on these transition slices, the primary fixed-threshold detector can fail to find any background at all (an apparent mask coverage near 100%), triggering a per-slice adaptive-Otsu re-estimate for that slice only. This fallback is not fully closed as an edge case, see Section 17 for the specific failure mode it can itself produce on very high-porosity content. For `sample_01–05`, the original circular-fit method (Section 9.3) is retained unchanged, since it already

performs correctly on these genuinely circular samples; sample_05 specifically requires a fixed intensity threshold rather than adaptive Otsu for its own boundary fit, because its unusually high porosity (Section 9.5, Table 19) leaves too few solid-material pixels for Otsu’s two-class assumption to separate correctly, a sample-content-dependent choice rather than a universal preference for one threshold strategy over the other.

14.6 A Documented Negative Result: Canny Edge Detection

Canny edge detection with morphological closing was investigated as a further alternative to both the circular-fit and shape-aware boundary methods, restricted to boundary detection only (it was not carried forward into any trained-model comparison). On sample_01–05, its results closely reproduce the production circular-fit numbers (Section 15.3), confirming those results independently rather than replacing them. On sample_06/07, it exhibits a specific and confirmed failure mode on a small number of slices: rather than following the true material boundary, it confidently traces the edge of the surrounding support scaffold instead, producing a boundary that looks well-formed rather than an obvious error. Investigation traced this to the underlying image signal itself: at the specific slices where the failure occurs, the material-to-background intensity contrast has genuinely faded to approximately the same strength as the scaffold’s own edge contrast, so no threshold, smoothing parameter, or shape heuristic (solidity, area continuity, and brightness were each tried as tie-breakers) can reliably distinguish the two closed shapes from the image content alone. This is reported as a negative result specifically because the cause is a property of the data at those depths rather than a fixable implementation choice; the shape-aware method of Section 14.5 remains the production method for sample_06/07 for this reason.

14.7 Background-Masking Correction for Detector Inference

A separate defect, found by inspecting the trained detector’s predictions on sample_07 (an out-of-distribution inference check, not part of the training or validation roles of Section 14.3), was that the network produced implausible defect predictions on slices with little or no metal in frame, near-empty transition and support-structure slices the training data never specifically taught it to recognise as background. The fix applies the same shape-aware boundary detector used in Section 14.5 to zero out background both before the trained network sees a slice and again on its predicted output, rather than relying on the network to infer the sample boundary implicitly from intensity alone.

14.8 Conv-Block Attribution Analysis

To identify which stages of the trained 2D U-Net (architecture per Section 3.7.1, verified by direct instantiation to have exactly 31,037,633 trainable parameters) contribute most to the defect decision, a forward hook is registered on every convolutional block’s output (four encoder blocks, the bottleneck, and four decoder blocks, nine blocks total), and

`retain_grad()` is called on each hooked output. A forward and backward pass is run on a representative 256×256 crop from `sample_02` (14.27% genuine Bernsen-labelled defect content), with the scalar objective $\sum(\hat{y} \odot y)$, the predicted defect probability map $\hat{y}$ elementwise-multiplied by the Bernsen ground-truth mask y and summed, backpropagated to populate every hooked block’s gradient. Each block’s relative contribution is then computed as the mean absolute value of activation times gradient, $|\overline{a \cdot \nabla a}|$, at that block, normalised across all nine blocks to sum to 100%. This is a single representative crop and a single attribution method, $\text{gradient} \times \text{activation}$ is a simple, interpretable choice but not the only one available, so the result reported in Section 15.6 is directional evidence about where the network’s decision-relevant signal concentrates, not an exhaustive architectural study. The analysis is run on the checkpoint identified in Section 14.3 as the strongest of the four loss-function configurations (BCE), verified to be the correct checkpoint by an exact match between its own recorded validation Dice and best epoch (0.7275 at epoch 13) and the same figures reported independently for the loss-ablation’s BCE run.

14.9 External Validation Against Independently Published Porosity

Otsu, Yen, and Bernsen thresholding are each compared, per NIST CoCr sample, against the porosity values independently published by Kim et al. [21] for the same physical samples, using the Pearson correlation coefficient r across the five samples in common. This validates the classical methods against a source external to this pipeline, rather than against another output of the same pipeline. A further check was made of Kim et al.’s own reference values themselves: their paper reports cross-validating their XCT-derived porosity against an independent gravimetric (density-based) method, computed from measured disk mass and cobalt-chrome’s nominal density, for several of their samples, with the two methods agreeing to within 1–4% absolute and an XCT-side threshold-sensitivity uncertainty below $\pm 0.12\%$ [21]. This means the reference this thesis’s own r value is measured against is not itself only another XCT measurement, it carries independent physical grounding one step back. This gravimetric cross-check in Kim et al.’s own paper was reported for specific samples (their own sample numbering 2, 4, 5, and 6); it has not been independently confirmed here that every one of the five NIST samples used in this thesis’s correlation individually received that exact check, as opposed to being reported as part of their broader dataset characterisation, and this caveat is carried into the results and limitations of this part accordingly.

14.10 Physical Voxel-Size Determination

`config.py`’s `PIXEL_SIZE_UM` constant, which converts pixel-based measurements elsewhere in this pipeline to physical units, was never calibrated to either dataset’s actual scan resolution and remains at its unfilled placeholder value of 1.0 (Section 12); this part reports the actual physical scale for both datasets as a measurement, without editing that constant in the running pipeline. For PODFAM, the raw TIFF files for `sample_06` and `sample_07` carry embedded `XResolution/YResolution` metadata tags, read directly from the file headers, giving a consistent voxel size of $12.001 \mu\text{m}/\text{px}$ for both samples. For the NIST CoCr samples, the raw TIFF files carry no equivalent spatial metadata (only an ImageJ-style intensity min/max in each file’s `ImageDescription` field), so the voxel size is

instead taken from the dataset’s own published documentation [5], which states a nominal voxel size of approximately 2.5 μm for most samples in the set, with samples 3 and 5 specifically scanned at a substantially finer 0.87 μm . This roughly threefold difference in resolution for exactly the two samples this thesis already excludes from training, sample_03 as near-empty and sample_05 for the DCT labelling artefact (Sections 9.4, 9.5), is noted in Section 16 as a plausible, previously unidentified explanation worth stating explicitly, a finer voxel size implies a smaller physical field of view per slice for the same pixel dimensions, which could plausibly affect both the amount of defect content captured and the image’s intensity/contrast profile. This is reported here as a hypothesis consistent with the available evidence, not as a confirmed causal finding; it has not been independently tested against the two samples’ actual field-of-view dimensions or intensity statistics within this work.

15. Results — Extended Validation

15.1 Loss-Function Ablation Results

Table 24 reports validation-set performance for all four loss configurations, trained and evaluated under the identical split described in Section 14.3.

Table 24: Loss-function ablation, validation-sample (sample_03) results. BCE achieves the best raw Dice and IoU of the four configurations.

Loss	Val Dice	Val IoU	Precision	Recall
BCE	0.7275	0.6235	0.809	0.7206
Dice	0.691	0.578	0.822	0.672
Combined Dice-Focal (Part II’s choice)	0.696	0.578	0.741	0.7278
Focal	0.574	0.466	0.976	0.471

BCE achieves the best Dice and IoU of the four configurations. It was not selected as the primary formulation when this comparison was first run, combined Dice-Focal was retained instead on the grounds of a more balanced precision/recall trade-off, but re-examining the exact figures shows that framing overstated the difference: BCE’s recall (0.7206) is only 0.0072 below combined Dice-Focal’s (0.7278), not a meaningful trade-off given the other three metrics all favour BCE outright. Separately, and independently of this ablation, the held-out generalisation-gap checkpoint identified in Section 15.2 below records a validation Dice of 0.7275 at epoch 13, an exact match to this table’s BCE row, confirming that checkpoint was already trained with BCE loss. Taken together, these two points of evidence support formalising BCE as the primary loss function for this pipeline going forward, rather than combined Dice-Focal.

15.2 The Generalisation Gap

Table 25 reports the same checkpoints’ Dice on their validation sample (sample_03) against their Dice on the fully held-out test sample (sample_01, never included in training or validation for any configuration in this part).

Every configuration shows the same pattern: validation Dice in the 0.70–0.73 range, held-out test Dice far lower, 0.12–0.21. This gap is present in the original (Part II) configuration as well (Section 10.3: validation Dice 0.600, held-out test Dice 0.147), so it is corroborated by two independently run experiments using different sample-role assignments and different loss/preprocessing configurations, not an artefact of one specific run. Table 26 reports the full held-out test metrics for the two directly comparable configurations (baseline vs. BHC/ring, circular mask both).

Table 25: Validation Dice vs. held-out test Dice, across the preprocessing and masking ablations of Sections 14.3–14.5. All configurations share the same held-out sample, sample_01. Acceptance criteria (Dice ≥ 0.75 , IoU ≥ 0.60 , Recall ≥ 0.80) fail in every configuration on the held-out sample.

Configuration	Validation Dice	Held-out test Dice
BCE, baseline (3-stage) preprocessing, circular mask	0.7275	0.1218
BCE, + BHC/ring preprocessing, circular mask	0.7013	0.1693
BCE, + BHC/ring preprocessing, shape-aware mask	~ 0.70	0.2119

Table 26: Preprocessing ablation, full held-out test set metrics (sample_01), circular mask both configurations.

Configuration	Dice	IoU	Precision	Recall
Baseline (3-stage preprocessing)	0.1218	0.0907	0.2908	0.3134
+ BHC and ring-artefact correction	0.1693	0.1328	0.4049	0.2974

Held-out test Dice improves by 39% relative ($0.1218 \rightarrow 0.1693$) with BHC/ring correction added, while validation Dice moves slightly in the opposite direction ($0.7275 \rightarrow 0.7013$). Read together, this indicates the preprocessing change improves genuine generalisation to unseen data rather than fitting the validation sample more closely, since the more trustworthy of the two numbers (held-out, never used for any training decision) improved while the less trustworthy one (validation, used for checkpoint selection) did not. Adding the shape-aware boundary mask of Section 14.5 on top of BHC/ring continues the same trend, held-out Dice rises further to 0.2119, a monotonic improvement across all three configurations ($0.1218 \rightarrow 0.1693 \rightarrow 0.2119$) even though every configuration still fails the thesis’s acceptance criteria by a wide margin.

15.3 Shape-Aware Masking Results

Table 27 reports the corrected full-volume Bernsen defect statistics for sample_06 and sample_07 using the shape-aware boundary method of Section 14.5, alongside the uncorrected result the original erosion setting produced.

Table 27: sample_06/07, shape-aware boundary detection, before and after erosion-margin correction. sample_06’s initial result was found to be a boundary artefact, not genuine porosity; sample_07’s corrected result was confirmed genuine by visual inspection of the full slice mosaic.

Sample	Defect fraction, erosion=8px (uncorrected)	Defect fraction, erosion=35px (corrected)
sample_06	2.92% (boundary rim artefact)	0.030%
sample_07	—	0.759%

sample_06’s uncorrected result (2.92%) is roughly two orders of magnitude above its corrected value (0.030%), traced to a thin rim of partial-volume boundary pixels being misclassified as defect at the original 8 px erosion margin; a sweep of erosion values (25, 30, 35, and 40 px) showed the false rim shrinking smoothly and predictably as the margin increased, with 35 px selected as the value fully removing the artefact without eroding genuine interior material. sample_07’s corrected result (0.759%) was cross-checked by direct visual inspection of its full slice mosaic and judged to reflect genuine porosity rather than a residual boundary effect. For comparison, the circular-fit method applied to sample_01–05 in this same later pipeline pass gives full-volume Bernsen defect fractions of 0.055% (sample_01), 6.494% (sample_02), 0.181% (sample_03), 1.641% (sample_04, adaptive Otsu boundary threshold), and 53.42% (sample_05, fixed boundary threshold, required because its $\sim 72\%$ porosity leaves too few solid pixels for Otsu’s two-class assumption to hold, Section 14.5). These figures are from an independently re-run pipeline pass and are not identical to Section 10.1’s Table 19 figures for the same samples, both are genuine pipeline outputs from different points in this work and are reported here as such rather than reconciled into one number.

The Pearson correlation between each classical method’s per-sample defect fraction and Kim et al.’s [21] independently published porosity (Section 14.9, five NIST samples in common) was recomputed under the circular, no-mask, and shape-aware boundary configurations, to check whether replacing the circular assumption changes the pipeline’s already-established external-validation accuracy. Table 28 shows the result.

Table 28: Pearson r vs. Kim et al.’s published porosity, by boundary-detection method (NIST samples, $n = 5$). Removing the boundary mask entirely is unusable; shape-aware masking is statistically indistinguishable from the circular mask it replaces.

Method	Circular mask	No mask	Shape-aware mask
Otsu	0.983	0.984	0.982
Yen	0.503	~ 0.43	~ 0.50
Bernsen	0.995	0.992	0.991

With no boundary mask at all, background air is misclassified as defect on low-porosity samples, Otsu’s defect fraction on one near-zero-porosity sample jumped from 0% to 24% once the mask was removed, making that configuration unusable despite its r value alone looking comparable. Shape-aware masking, by contrast, reproduces the circular mask’s accuracy to within 0.001–0.004 of r across all three methods, recovering the same measured accuracy with a method that carries no assumption about cross-sectional shape.

15.4 Background-Masking Fix for Detector Inference

Table 29 reports the trained detector’s predicted defect fraction on sample_07 (out-of-distribution inference only, Section 14.7) before and after applying the shape-aware background-masking fix.

The fix produces a $5.6\times$ reduction ($26.778\% \rightarrow 4.764\%$) using the identical checkpoint on the identical sample, only the background masking changed. This confirms most of the prefix number was background hallucination on transition/support-structure content rather than a fundamental defect- detection failure. sample_07 was never in this checkpoint’s

Table 29: sample_07, U-Net predicted defect fraction, same checkpoint and same sample, before and after the background-masking fix.

Configuration	Predicted defect fraction
2.5D U-Net, before fix	26.778%
2.5D U-Net, after fix	4.764%
2D U-Net, after fix	2.691%
Bernsen classical (reference, same sample)	7.060%

training distribution (different geometry, and processed with a different preprocessing configuration than the training samples), so none of the figures in this table can be checked against ground truth; they are reported as an internal consistency check on the fix, not as a validated accuracy result.

15.5 Canny Boundary Detection: Validation and Failure

On sample_01–05, Canny edge detection with morphological closing reproduces the production circular-fit defect fractions closely, 6.62% vs. 6.49% for sample_02, and 1.69% vs. 1.64% for sample_04, cross-confirming those production numbers by an independent method. On sample_06/07, it correctly follows the true material boundary through most of each volume’s depth, but on two slices it instead traces the surrounding support scaffold, reporting 29.96% defect content against a true value of approximately 0–2% at those slices. As discussed in Section 14.6, this failure was traced to the underlying image content (a genuinely ambiguous material/scaffold contrast at those specific slices) rather than a fixable parameter, and the shape-aware method of Section 14.5 remains the production choice for these two samples for that reason.

15.6 Conv-Block Attribution Results

Table 30 reports each convolutional block’s relative contribution to the defect decision, computed as described in Section 14.8, on the BCE checkpoint identified in Section 15.1 as the strongest of the four loss-function configurations.

encoder_1 through encoder_3 together account for 68.88% of the attributed signal, while decoder_4, the block closest to the final output layer, is the least influential of all nine (1.28%). This analysis was first run, in error, on the combined Dice-Focal checkpoint (`top6_by_voids`) rather than the correct BCE checkpoint, before the ablation of Section 15.1 had settled which checkpoint should be treated as primary; that run found the same qualitative pattern at a smaller magnitude (encoder_1–3 combined: 61.7%; decoder_4: 1.46%). Re-running the analysis on the verified BCE checkpoint after the fact, the pattern reported in Table 30 replicated cleanly and was, if anything, stronger.

Table 30: Gradient $\times$ activation attribution per convolutional block, BCE checkpoint (verified val_dice = 0.7275, epoch 13). The three encoder blocks together account for approximately 69% of the total attributed signal; the final decoder block is the least influential of all nine.

Block	Channels	Attributed signal
encoder_1	64	27.78%
encoder_2	128	25.81%
encoder_3	256	15.29%
encoder_4	512	6.84%
decoder_3	128	6.36%
decoder_2	256	6.10%
bottleneck	1024	5.92%
decoder_1	512	4.63%
decoder_4	64	1.28%

15.7 External Validation and Voxel-Size Results

The gravimetric cross-check identified in Section 14.9 covers, in Kim et al.’s own sample numbering, samples 2, 4, 5, and 6, with XCT-vs-gravimetric agreement within 1–4% absolute and threshold- sensitivity uncertainty below $\pm 0.12\%$ [21]. This strengthens the interpretation of this thesis’s own $r = 0.995$ Bernsen-vs-Kim-et-al. correlation, Table 28, as agreement with a reference value that already carries independent physical grounding rather than a purely XCT-internal comparison, with the caveat, stated in Section 14.9, that individual per-sample confirmation of the gravimetric check for all five NIST samples used in this thesis’s own correlation was not independently verified here.

The physical voxel-size figures determined in Section 14.10 are: PODFAM sample_06 and sample_07, 12.001 $\mu\text{m}/\text{px}$ (from embedded raw-TIFF resolution meta-data, consistent across both samples), giving approximate physical dimensions of 12.3 mm $\times$ 14.8 mm $\times$ 17.6 mm for sample_06 and 14.6 mm $\times$ 13.7 mm $\times$ 17.7 mm for sample_07; and NIST CoCr samples, approximately 2.5 $\mu\text{m}/\text{px}$ for samples 1, 2, and 4, and 0.87 $\mu\text{m}/\text{px}$ for samples 3 and 5, per the dataset’s own published documentation [5]. `config.py`’s `PIXEL_SIZE_UM` constant itself remains at its unfilled placeholder value in the running pipeline; these figures are reported as an external measurement against the raw data and documentation, not as a change already applied to the pipeline’s own unit conversions.

16. Discussion — Extended Validation

The single most important result in this part is the generalisation gap itself (Section 15.2), not any one preprocessing or masking choice. Validation Dice in the 0.60–0.73 range was already established as a plausible, defensible detection result by Part II; this part shows that number does not predict performance on a sample genuinely never seen during training, which instead falls to 0.12–0.21 across every configuration tested, including the original Part II configuration itself (0.600 vs. 0.147, Section 10.3). Because this pattern reproduces across two independently run experiments with different sample-role assignments, different loss functions, and different preprocessing pipelines, it is best explained as a property of this pipeline’s train/validation/test design, specifically, how few distinct samples are available to fill the validation and held-out-test roles, rather than as an artefact of any single training run. This reframes the acceptance-criteria shortfall already stated in Part II (Section 10.2) as a structural finding rather than a training issue that more epochs or a different loss function would resolve on its own.

Within that constraint, the preprocessing and masking ablations (Sections 15.2, 15.3) still show a genuine, monotonic improvement in the more trustworthy of the two numbers: held-out test Dice rises from 0.1218 to 0.1693 to 0.2119 as BHC/ring correction and then shape-aware masking are added, even as validation Dice stays flat or drifts slightly downward. Read together with the generalisation-gap finding above, this supports two separate conclusions rather than one: the pipeline’s absolute detection accuracy on unseen data remains far below the acceptance criteria, and the specific preprocessing and masking changes tested here are moving that number in the right direction rather than the wrong one, evidence that the improvements are real generalisation gains and not artefacts of fitting the validation sample more closely.

The loss-function ablation (Section 15.1) resolves an inconsistency in how Part II’s own headline result and its stated primary loss function relate to one another: the held-out generalisation-gap checkpoint’s own recorded validation Dice (0.7275, epoch 13) is an exact match to the loss-ablation’s BCE row, meaning the thesis’s central generalisation finding was already produced using BCE, not combined Dice-Focal, the loss formulation Part II states as primary. Combined with BCE’s higher raw Dice and IoU and a recall difference against Dice-Focal (0.0072) too small to support the earlier precision/recall trade-off argument for preferring Dice-Focal, this is read here as sufficient grounds to recommend BCE as the pipeline’s primary loss function, correcting rather than merely supplementing Part II’s stated choice.

The conv-block attribution result (Section 15.6) offers one concrete, if bounded, architectural implication: with `encoder_1–3` together responsible for roughly two-thirds of the network’s attributed defect-discriminating signal and the final decoder block the least influential of all nine, the highest-leverage architectural change for this task is more likely

to be strengthening the early encoder stages or their skip connections, for instance via attention gating, than adding depth to the bottleneck or decoder. This is offered as a direction for future architecture work rather than a finding validated by ablation; a single attribution method on a single representative crop is suggestive evidence of where signal concentrates, not proof that modifying those specific blocks would improve held-out performance.

The boundary-detection work (Sections 15.3–15.5) illustrates a pattern already established once in this thesis, in Section 9.5/10.5’s diagnosis of Bernsen’s DCT-calibration failure, and repeated here in a different subsystem: a method that produces a superficially plausible number can still be wrong by a large factor, and the wrongness is only caught by an independent cross-check. `sample_06`’s 2.92% uncorrected defect fraction was a physically plausible-looking value for a metal AM part; only the erosion-margin sweep revealed it was a boundary artefact rather than genuine porosity. Canny edge detection (Section 15.5) supplies the same lesson from the opposite direction: it independently confirms the circular-fit method’s numbers on samples where both methods work, which is useful corroboration, but it was not adopted for `sample_06/07` despite otherwise working correctly through most of each volume’s depth, because a method that is right most of the time but silently wrong in a way that looks correct is a worse production choice than a method validated not to exhibit that failure mode at all.

Finally, the two findings of Sections 14.9–14.10 and 14.10 respectively strengthen and complicate the thesis’s use of the NIST CoCr dataset. The gravimetric cross-check in Kim et al.’s own paper means the $r = 0.995$ external-validation figure (Table 28) is agreement with a reference that already carries independent physical grounding, a stronger claim than XCT-vs-XCT agreement alone would support, even with the caveat that individual per-sample confirmation of that check was not verified here. Conversely, the voxel-size finding, samples 3 and 5 scanned at roughly a third of the other samples’ voxel size, raises a plausible, previously unconsidered explanation for why those specific two samples were the ones excluded from this thesis’s own training and ranking (Sections 9.4, 9.5): a substantially finer scan resolution implies a smaller physical field of view per slice at the same pixel dimensions, which could independently affect both defect content captured and intensity/contrast statistics. This is stated explicitly as an unconfirmed hypothesis rather than a demonstrated cause, consistent with this thesis’s practice elsewhere of separating what has been measured from what remains plausible but unverified, but it is a concrete, checkable direction for follow-up work that this part’s own investigation did not have scope to pursue further.

17. Limitations — Extended Validation

This section states the limitations specific to the work reported in Sections 14–16. It supersedes or narrows two of Part II’s own stated limitations (Section 12) for the specific samples and checkpoints this part covers; those two points are noted explicitly below rather than left for the reader to reconcile.

- **The generalisation gap remains unresolved, not just measured.** This part characterises the gap between validation and held-out test performance across several configurations (Section 15.2) but does not close it. The highest-leverage untried fix, retraining on a larger, rotating pool of samples rather than the fixed four-sample, single-role split used throughout this thesis, was identified but not attempted within this work’s scope.
- **Every configuration in this part shares the same single held-out test sample, sample_01.** A generalisation-gap finding replicated across preprocessing and loss-function variations on one held-out sample is stronger evidence than a single run, but it is still one sample; it cannot rule out that a different held-out sample would show a smaller, or larger, gap.
- **The shape-aware boundary method’s transition-slice fallback (Section 14.5) is not fully hardened.** The per-slice adaptive-Otsu fallback triggered on transition slices near a volume’s top or bottom can itself overcorrect on very-high-porosity content, sample_05 at $\sim 72\%$ porosity, or on some of sample_06’s own transition slices, producing an implausibly low coverage estimate (1–5%) in the opposite direction from the failure it was designed to catch. The full seven-sample production run completed with no unresolved failures under the current hybrid logic, but this is an asymmetric, only partially closed edge case rather than a solved one.
- **The conv-block attribution result (Section 15.6) uses one representative crop and one attribution method.** It is directional evidence about where the trained network’s decision-relevant signal concentrates, not a validated architectural claim; a different crop, sample, or attribution method could in principle produce a different relative ranking among the nine blocks.
- **The gravimetric cross-check strengthening the external-validation claim (Sections 14.9, 15.7) is reported at the level of Kim et al.’s paper as a whole, not confirmed per sample.** Their gravimetric comparison covers specific samples in their own numbering; it has not been independently verified here that every one of the five NIST samples used in this thesis’s own $r = 0.995$ correlation

individually received that exact cross-check, as opposed to being reported as part of a broader dataset characterisation.

- **The NIST voxel-size finding (Section 14.10) is an unconfirmed hypothesis, not a demonstrated cause.** That samples 3 and 5 were scanned at a substantially finer resolution than the other NIST samples, and are also the two samples this thesis excludes from training, is reported as a plausible explanation worth investigating, not as a verified causal link; this part did not test the two excluded samples' actual field-of-view dimensions or intensity statistics against this hypothesis.
- **This part's own PODFAM defect-fraction figures for sample_01–05 (Table 27 discussion, Section 15.3) come from a later, independently re-run pipeline pass and differ numerically from Section 10.1's Table 19 figures for the same samples.** Both are genuine outputs of the working pipeline at different points in this work; they are reported here as such, rather than reconciled into a single authoritative number, consistent with this thesis's practice elsewhere of not silently merging figures from different runs.
- **Supersedes Part II, Section 12, PIXEL_SIZE_UM bullet, for sample_06/07 specifically:** the physical voxel size for these two samples is now known (12.001 $\mu\text{m}/\text{px}$, Section 14.10), resolving Part II's stated uncertainty for this pair. It remains unresolved for sample_01–05 (see the NIST voxel-size figures above, which come from the dataset's published documentation rather than an in-house measurement) and `config.py`'s `PIXEL_SIZE_UM` constant itself has not been updated to reflect either figure, so any pipeline output still expressed in that constant's units remains uncalibrated.
- **Narrows Part II, Section 12, sample_07-single-data-point bullet:** sample_07 is now supported by a second, independent line of evidence beyond the original single inference-only check, the background-masking fix's internal consistency check (Section 15.4) and the shape-aware boundary method's corrected, visually-confirmed defect fraction (Section 15.3), but it still has no ground-truth role and no retraining took place; Part II's original caveat that its results are a plausibility check rather than a validated accuracy figure still applies.

18. Conclusion

This part revisited three design choices made without controlled comparison earlier in the thesis, loss function, preprocessing, and sample-boundary detection, and reported two further analyses, network interpretability and a strengthened external validation, on the same PODFAM pipeline validated in Part II. A controlled four-way loss-function comparison found BCE, not the combined Dice-Focal formulation stated as primary in Part II, achieves the strongest validation Dice and IoU, and further established that Part II’s own headline generalisation result was already trained with BCE, correcting rather than merely refining Part II’s stated choice. A held-out test sample, evaluated across every configuration tested in this part, consistently scored far below its corresponding validation sample, 0.12–0.21 Dice against 0.70–0.73, replicating, on an independent split and set of configurations, the same pattern Part II first reported for its own single trained model (Section 10.3). This generalisation gap, not any single preprocessing or masking result, is this part’s central finding: it is a property of how few distinct samples this pipeline has to work with, not of any one training run, and it remains the pipeline’s primary open problem going into any future work on this dataset.

Within that constraint, beam-hardening/ring-artefact correction and a shape-agnostic replacement for the circular sample-boundary assumption each moved held-out test Dice in the right direction ($0.1218 \rightarrow 0.1693 \rightarrow 0.2119$), evidence that these specific engineering changes improve genuine generalisation rather than only fitting the validation sample more closely. The shape-aware boundary method also corrected a two-order-of-magnitude artefact in sample_06’s reported defect fraction ($2.92\% \rightarrow 0.030\%$) and recovered the same external-validation accuracy as the circular mask it replaced (Table 28) while removing an assumption already known not to hold for two of the seven PODFAM samples. A parallel investigation of Canny edge detection as a further alternative is reported as a negative result: it independently corroborates the circular-fit method on samples where both apply, but exhibits a confirmed, data-driven failure mode on the non-circular samples that the adopted shape-aware method does not, and is documented here specifically so that choice is not revisited without this evidence.

A gradient-based attribution analysis of the trained detector, run on the verified BCE checkpoint, found that the three encoder blocks together account for roughly two-thirds of the network’s attributed defect-discriminating signal, with the block closest to the output the least influential of all nine, offering a concrete, evidence-based direction (strengthening early-encoder feature extraction or its skip connections) for future architecture work on this task. Separately, checking Kim et al.’s own reference porosity values against their paper’s stated methodology found they were themselves cross-validated against an independent gravimetric method, strengthening the interpretation of this thesis’s $r = 0.995$ external-validation result as agreement with an already physically-grounded reference rather than an XCT-internal comparison alone. A further, still-unconfirmed observation, that the two NIST samples this thesis excludes from training were scanned at a substantially finer voxel

size than the rest of the set, is reported as a plausible explanation worth testing directly in any continuation of this work, rather than as a settled finding.

Taken together with Parts I and II, this document now covers method development on a public benchmark (NIST CoCr, Part I), full-volume classical-versus-deep-learning validation on real industrial process-monitoring data (PODFAM, Part II), and a controlled revisiting of that validation’s own design choices alongside two further, independent lines of evidence, interpretability and external-reference verification (Part III). The overall picture that emerges is consistent across all three parts: the pipeline’s classical thresholding stage, particularly Bernsen local adaptive thresholding, is validated against independently published, physically-grounded reference data with high confidence; the trained detector built on top of it produces plausible, well-motivated predictions within the sample or samples it was trained and validated on, but its performance on data it has never seen falls far short of this thesis’s stated acceptance criteria in every configuration tested; and every major result in this document, favourable or not, was reached only after an explicit, reported cross-check, itself a recurring finding of this work, that a classical method’s or a trained model’s output looking plausible in isolation is not sufficient grounds to trust it without independent verification.